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Phylogenetic Diversity and Endemism of the Vascular Flora of the Páramo Ecoregion

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Abstract—The páramo ecoregion is known for its remarkable biodiversity and evolutionary histories shaped by complex biogeographical processes. To protect and conserve the ecological diversity and evolutionary lineages in this hotspot requires analyzing spatial patterns of phylogenetic diversity and endemism. This study uses spatial phylogenetic methods, combining georeferenced occurrence data with a genus-level phylogeny of the region's entire vascular flora to investigate the spatial distributions of phylogenetic diversity and endemism. In this study, we assembled a comprehensive dataset of 361 equal-area grid cells covering the páramo biome and compiled occurrence records for 567 vascular plant genera. A phylogeny of vascular plants in the páramo was generated by combining publicly available genetic sequence data with novel genetic data from Colombian herbarium collections. This study examined the vascular plant diversity in the páramo ecosystem and identified distinct centers of neo- and paleoendemism, as well as a strong pattern of phylogenetic clustering across most páramo regions. The phylogenetic beta diversity analysis identified several phylogenetically distinct bioregions, characterized by both widespread, shared evolutionary histories and localized, distinct assemblages. This study provides a scope to prioritize certain areas for protection and to further identify and understand biogeographical patterns in the páramo ecoregion.

Keywords—Andes, biodiversity, conservation, neotropics, spatial phylogenetics.

Biodiversity is increasingly under pressure as humanity's dependence on ecosystem services intensifies, which creates an urgent need to maintain balanced ecosystem functions. Global biodiversity is undergoing a severe decline, driven by overexploitation, habitat conversion, invasive species, and climate change (IPBES 2019). This rapid loss, often referred to as the sixth mass extinction (Ceballos et al. 2020), underscores the urgent need for tools that can both effectively measure biodiversity and guide conservation efforts. Biodiversity encompasses all the variation of life across all levels, from genes to ecosystems (Purvis and Hector 2000), underpinning ecological and evolutionary processes that sustain life on Earth. The connection between these ecosystem services and the maintenance of balanced ecological processes is a key concern in today's world. Due to recent innovations and the pace of advancing technology, biodiversity measurement has changed substantially in the last few decades. Conservation planning driven by molecular data has received attention because of the recent advancements facilitating the acquisition of large datasets (Nielsen et al. 2023). Analyzing the spatial patterns of biodiversity has become important for improving conservation planning and management. The spatial phylogenetic approach requires the integration of species distribution data with phylogenetic information. Such analysis is possible now based on large databases such as GenBank (Benson et al. 2018) for molecular sequences and the Global Biodiversity Information Facility (GBIF) (GBIF 2024) for geospatial occurrence records.

Traditional biodiversity measures often prioritize quantifying species richness for their conservation strategies, yet Mishler (2023) described species as an arbitrary level of the tree of life. Species, as distinct, identifiable entities, can not fully describe the divergence of lineages. Spatial phylogenetics integrates evolutionary history with geographic distribution to provide a multidimensional view of biodiversity (Thornhill et al. 2016; Scherson et al. 2017). Spatial phylogenetics captures the spatial variation in lineage diversity

shaped by speciation, extinction, dispersal, and ecological filtering, while also reflecting past climatic and geological processes (Mishler et al. 2014; Thornhill et al. 2016; Scherson et al. 2017; Spalink et al. 2018). Spatial phylogenetics can be applied to any taxonomic rank, providing flexibility to conduct studies in any clades that represent monophyletic groups. This study employed a spatial phylogenetics approach to identify the spatial variation in the evolutionary diversity of páramo vascular plants for the first time. This incorporates a molecular phylogeny of páramo flora and describes the regional phylogenetic diversity, biogeographical patterns, and distinct floristic bioregions of the páramo.

The páramo ecoregion has high significance in terms of biodiversity and conservation. Located within the Tropical Andes, one of the world's most prominent biodiversity hotspots, it harbors a unique flora characterized by exceptional species richness and high levels of endemism (Luteyn 1999; Rangel 2000; Sklenář et al. 2011; Madriñán et al. 2013; Pérez-Escobar et al. 2022). Often described as a “hotspot within a hotspot” (Myers et al. 2000; Barthlott et al. 2007; Madriñán et al. 2013), the páramo is considered one of the world's “hyper-hot” biodiversity regions, where the majority of species are endemic (Madriñán et al. 2013; Londoño et al. 2014). The remarkable diversity of the páramo flora is shaped by the region's complex topography and heterogeneous climatic and historical conditions, which vary across elevations and landscapes (Flantua et al. 2019; Pérez-Escobar et al. 2022). This variability supports a wide array of taxonomic groups, including angiosperms, ferns, mosses, and liverworts. The flora and fauna have evolved to thrive under cold, high-altitude conditions with substantial moisture. Compared to other high-elevation ecosystems worldwide, such as the Afroalpine zone and the highlands of New Guinea, the páramo stands out for its particularly high species richness (Sklenář et al. 2011). Diversity patterns within the páramo are characterized by its geological history, rapid diversification

processes, and environmental conditions. The páramo region faces multiple, interrelated threats driven by both climate change and human activities, placing its exceptional biodiversity and vital ecosystem services at severe risk (Peyre et al. 2018; Diazgranados et al. 2021). Climatic projections predict a substantial decline in plant species and local assemblages and indicate an extinction risk for a significant fraction of the flora (Peyre et al. 2020). At the same time, anthropogenic disturbance further intensifies vulnerability, particularly at lower elevations where land-use practices such as grazing, burning, and cultivation strongly influence floristic composition (Valencia et al. 2012; Vásquez et al. 2015).

This high-mountain ecosystem's unique environmental and geographical conditions have influenced its flora over millions of years (Sklénár et al. 2014). Hughes and Eastwood (2006) described it as island radiation on a continental scale, which underwent an exceptional rate of plant diversification with Andean mountain uplift. The diversification of the páramo was influenced by Pleistocene glacial oscillations and niche shifts, which created ecological opportunities and diversification processes (Londoño et al. 2014; Kolář et al. 2016). These biogeographical events have fostered both the survival of ancient lineages and more recent speciation events, resulting in a rich and uniquely biodiverse páramo ecoregion. This makes it an appealing study system for spatial phylogenetic analysis.

The development of spatial phylogenetics stems from the foundational concept of phylogenetic diversity (PD), introduced by Faith (1992). The widely used metric PD is calculated as the sum of all the branch lengths of a phylogeny within a set of taxa, representing the total amount of evolutionary history (Scherson et al. 2017). The concept of endemism measures the portions of geographically rare biodiversity in a particular area. Faith et al. (2004) developed the concept of absolute phylogenetic endemism, which identifies the particular portion of the phylogenetic tree that is restricted to that region. Following this, the concept of phylogenetic endemism (PE) was developed by Rosauer et al. (2009) for identifying the spatial restriction of phylogenetic history by calculating mean geographic rarity for those taxa. Further advancements in spatial phylogenetics introduced two new metrics: relative phylogenetic diversity (RPD) and relative phylogenetic endemism (RPE) by Mishler et al. (2014). These derived metrics, RPD and RPE, were applied to distinguish different types of endemism that differentiate regions containing range-restricted short branches (neo-endemism) and range-restricted long branches (paleo-endemism). This method is the central concept of spatial phylogenetic analysis, named categorical analysis of paleo- and neo-endemism (CANAPE), which integrates geospatial datasets and molecular phylogenetic data. Finally, beta-diversity analysis based on phylogenetic diversity identifies beta diversity turnover (Graham and Fine 2008), which can distinguish communities based on their shared and unshared phylogenetic traits. The potential of this is to understand how communities differ based on their distinct evolutionary histories and environmental niches and to determine biogeographic regions.

The main aim of this study is to identify páramo regions that are of high conservation value, focusing on detecting exceptionally endemic areas within the already highly endemic North Andean páramo. This approach was expected to reveal highly concentrated endemism with distinct spatial

patterns shaped by the region's complex interplay of historical isolation, ecological opportunity, and subsequent diversification. Since the Andean páramo is considered a hotspot within a biodiversity hotspot, searching for further signals of significantly high endemism within this already highly endemic region may yield unexpected patterns. This study is the first to include all vascular plants from the Andean páramo ecoregion in a comprehensive phylogeny, which is then integrated with georeferenced occurrence records to reveal spatial patterns of phylogenetic diversity and endemism. The study also identifies bioregions based on shared lineages to distinguish floristically distinct regions. Finally, it provides regional priorities for conservation, offering evolutionarily informed recommendations for conservation prioritization.

MATERIALS AND METHODS

The spatial phylogenetic analysis was conducted first by defining the study region and acquiring georeferenced occurrence data. Taxon information was compiled, and molecular sequence data were retrieved to construct a comprehensive phylogenetic tree. Missing sequence data were obtained from specimens collected from the study region, followed by genome sequencing in the lab, and with these data a complete phylogeny was built. The resulting phylogeny was then integrated with georeferenced occurrence data. Finally, spatial diversity metrics were calculated and mapped across the region.

Study Region—The Andean páramo ecosystem spans the Andes Mountains in Colombia, Ecuador, Venezuela, and Peru, forming part of the World Wildlife Foundation (WWF) biome entitled 'Montane Grasslands and Shrublands' (Olson and Dinerstein 2002). WWF defines and groups these ecoregions based on their similar biogeographic characteristics to facilitate conservation planning and management (Olson et al. 2001). This study included four distinct terrestrial ecoregions: the Northern Andean Páramo (spanning Venezuela, Colombia, and Ecuador), the Santa Marta Páramo (Colombia), the Cordillera de Mérida Páramo (Venezuela), and the Cordillera Central Páramo (Ecuador and Peru) (Fig. 1) (Olson and Dinerstein 2002). The Cordillera Central Páramo is characterized extensively by the Huancabamba Depression in northern Peru, often described as a major biogeographic barrier (Anthelme et al. 2014). The páramo ecosystem has been the subject of various biogeographic regionalization efforts, resulting in a consensus around a single páramo biogeographic zone (Jiménez-Rivillas et al. 2018; Peyre et al. 2018; Acevedo et al. 2020). Subsequent work proposed a more comprehensive páramo biogeographical unit, which together constitute a páramo province within the South American Transition Zone (Morrone et al. 2022). This consensus does not completely align with WWF-defined ecoregions; in particular, the Cordillera Central Páramo in Peru was excluded from studies by Morrone et al. (2022) due to its distinct bioclimatic characteristics. The biogeographic barrier with northern Peru is considered a transition zone between the Páramo and the Puna phytogeographic units (Cuesta et al. 2017). Although northern Peru has distinct vegetation characteristics, it was retained in this study to ensure latitudinal continuity along the Andean chain, and because comparative studies have highlighted floristic and evolutionary affinities between Peruvian and northern páramos (Lutelyn 1999; Cuesta et al. 2017). Additionally, it aligns with the WWF Global 200 Ecoregions framework, which includes the Cordillera Central páramos in its páramo ecoregion delineation and provides a single páramo polygon layer. For practical reasons, this study followed the páramo ecoregion delimited by WWF. The polygon data were downloaded from the WWF database "WWF's Global 200 project" (worldwildlife.org/publications/global-200).

Taxonomic Selection and Occurrence Data—The Global Biodiversity Information Facility (GBIF) database contains occurrence records which were selected from within the polygon for the northern Andean páramo ecoregion. Based on the availability of genetic sequence and occurrence records, this study was conducted at the taxonomic level of genus, following the lead of previous studies (Heenan et al. 2017; Scherson et al. 2017; Cheikh Albassatneh et al. 2021). The taxonomic list was derived from GBIF, and taxonomic accuracy was assessed with the GBIF Backbone Taxonomy (GBIF 2023). Additionally, species names were cross-referenced with National Center for Biotechnology Information (NCBI) taxon identifiers to retrieve sequence data, thereby avoiding the issues related to taxonomic mismatches and synonyms. The taxonomic list obtained from the GBIF

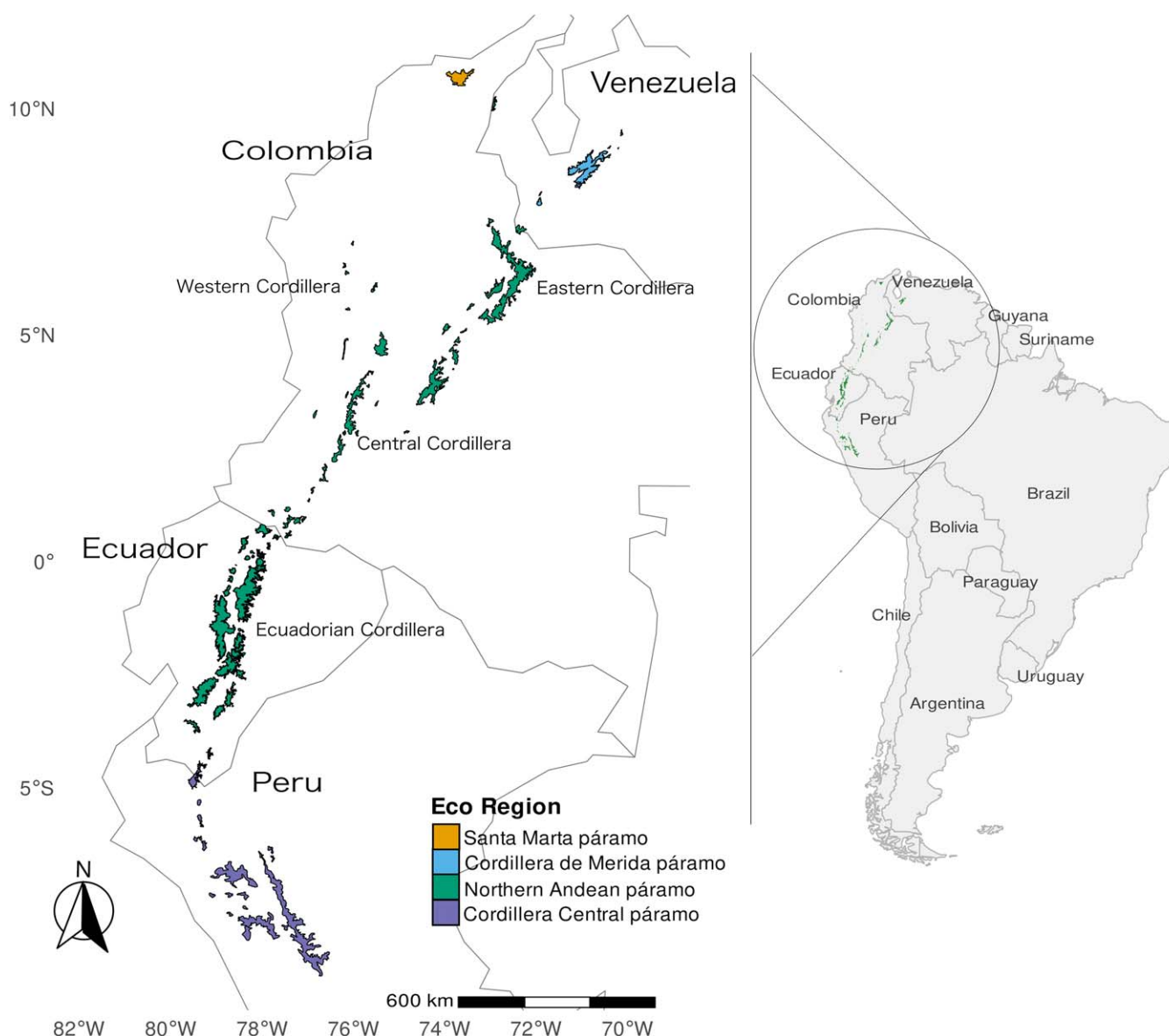


FIG. 1. Detailed map of the páramo, representing WWF's four ecoregions. To enable discussion of various subregions of the Northern Andean páramo, four 'cordillera' demarcations are highlighted (Luteyn 1999).

occurrence records was used for all spatial phylogenetic analyses. Occurrence records were retrieved from GBIF using the R package "rgbif" (Chamberlain et al. 2025) with the taxon key "Tracheophyta" and specifying the páramo polygon extent as a well-known text (WKT). The WKT served as a spatial filter capturing all occurrence records within the páramo polygon layer (GBIF 2024).

This polygon layer was created by merging the páramo ecoregions Northern Andean páramo, Cordillera de Merida páramo, Santa Marta páramo, and Cordillera Central páramo. All spatial data was transformed to the appropriate coordinate reference system (CRS) using the Universal Transverse Mercator (UTM) projection (epsg:32618) for Colombia to ensure accurate spatial analysis. After obtaining the occurrence dataset, the R package "sf" (Pebesma 2018) function "st_intersection" was used to remove records that were outside the páramo geographic layer. This process limited the occurrence records obtained from GBIF to the páramo polygon and also excluded any records outside of the polygon. This initial taxonomic selection identified a total of 1833 genera within the WWF-specified ecoregion layer, which were chosen for constructing the phylogeny without extensive data cleaning. This taxonomic list was selected for building the phylogeny, with each genus representing an operational taxonomic unit (OTU).

A multi-step data cleaning process was followed to ensure the quality of the occurrence records. The R package "CoordinateCleaner" was

utilized to eliminate duplicate and flagged records, including those with potential coordinate errors, such as records near administrative centroids, urban areas, or institutions (Zizka et al. 2019). Next, the records with coordinate uncertainty greater than 10 km were eliminated, while retaining those with unspecified uncertainty, and limited the records that were collected after 1960. After completion of this data cleaning process, the taxonomic list was filtered based on a native plant list that excluded non-native plants, invasive, and crop plants. The native plant list was obtained from the Andean páramo flora database, Vegparamo (Peyre et al. 2015), and from the Colombian Biotic Diversity XVI dataset (Churio and Orlando 2018). This process identified 427 unique genera in the vegparamo dataset and 573 genera in Colombian Biotic Diversity dataset. Together, the datasets comprise a total of 683 unique genera, with 317 genera common to both. From the 683 genera identified in both the Vegparamo and the Colombian páramo plant data, 567 matched our GBIF obtained genera. Those 567 genera account for a total of 119,909 occurrence records, which were selected for analyzing all diversity metrics (Table S2).

The distribution of biodiversity metrics can vary based on the observation scale. Smaller scales can capture detailed patterns in local areas, while larger scales are better for detecting broader patterns. Biodiversity metrics like species richness, endemism, and functional diversity can also differ at different observation scales. To determine the optimal resolution this study conducted several tests using grid sizes of 5 km, 10 km, 15 km,

20 km, and 25 km on the páramo polygon layer. Spatial community data was generated using the “points2comm” function in R “phyloregion” package (Daru et al. 2020), converting occurrence records into a community data matrix for all possible grid resolutions. Each grid resolution’s occurrences were converted into a presence-absence matrix, which was then used to compute redundancy values representing the proportion of species redundancy within each grid. The matrix indicates species presence or absence based on recorded specimens, and the redundancy value for a given cell was calculated using the formula $(1 - R/N)$, where R is the species richness within the cell and N is the number of occurrences within the cell) (Garcillán et al. 2003; Nitta et al. 2022).

As páramo ecosystems are unique environments located around 3000 meters in elevation, they often resemble isolated ‘high-mountain islands’ (Kolář et al. 2016). For spatial analyses in such complex geographies, it was essential to verify the completeness of the grid cells to ensure that each accurately reflects the geographic conditions and features found within the páramo. While grid cells are defined with equal overall area, the actual land area within them is not uniformly distributed, particularly in areas with fragmented or complex topographies, which can influence sampling. Therefore, instead of removing occurrence cells based on comparisons with land area, this study filtered out undersampled cells by analysing completeness values, following a study on Japan’s coastal island geography (Nitta et al. 2022). The present study excluded grid cells with a redundancy value of less than 0.1 which reduced the risk of significant data loss from strictly filtering cells with incomplete land coverage and ensured the retained cells had adequate sampling density.

DNA Sequence Collection and Generation via Herbarium Genomics—In order to create a phylogenetic tree containing all the taxa found in the páramo region, representative sequence data were sourced from the public repository NCBI GenBank (Benson et al. 2018). For this, the highly conserved chloroplast-genome maturase K gene (*matK*), chloroplast NADH dehydrogenase F gene (*ndhF*), RuBisCO large subunit gene (*rbcL*), and *trnF*(GAA) gene, as well as the internal transcribed spacer of the nuclear ribosomal DNA (*ITS*) were selected. From the initial list of 1833 taxa, available sequences from at least one of the selected markers were found for about 1670 genera. These sequences were identified and downloaded from NCBI using the Python script Matrix Maker (Freyman and Thornhill 2016). If at least one species of a genus had a publicly available sequence for the marker in question, the sequence was included in the study to represent its genus.

To supplement this set of DNA sequences, leaf tissue samples were collected from 33 dried herbarium plant specimens in the Herbario Luis Sigüfredo Espinal Tascón (CUVC) at the University of Valle in Cali, Colombia. Genomic DNA was extracted from these leaf tissue samples using the Qiagen DNeasy Plant Mini Kit, following the manufacturer’s protocol. After DNA extraction, the concentration of the extracted DNA was measured using the Qubit 4.0 Fluorometer (Life Technologies) with the dsDNA HS Assay Kit. Following extraction, DNA samples were sheared to the desired fragment size (400–600 bp) using a Covaris ME220 focused ultrasonicator. The sheared genomic DNA extracts were used to construct Illumina sequencing libraries following the BEST (blunt-end single-tube) library preparation protocol (Carøe et al. 2018), which involves three main steps: blunt-end repair, adapter ligation, and adapter fill-in. After a qPCR analysis to determine the optimal number of cycles, the purified libraries were subjected to dual-indexing PCR with 16–30 cycles. Each PCR reaction was carried out in a 100-μL volume containing: 7.5 μL of library template, 0.8 μL of dNTPs (25 mM), 2 μL each of F and R index primers (10 μM), 1 μL of AmpliTaq Gold polymerase (5 U/μL), 10 μL of AmpliTaq Gold buffer (10X), 10 μL of MgCl₂ (25 mM), 2 μL of BSA (20 mg/mL), and 64.7 μL of H₂O. The resulting libraries were purified and size-selected using a 1:1 ratio of purified DNA:SPRI beads, and then assayed and quantified on the Agilent 4200 TapeStation using HS D1000 ScreenTapes and reagents, prior to equimolar pooling. Sequencing was performed by the Novogene UK commercial sequencing service on the NovaSeq X sequencing platform (150-bp paired end chemistry).

The raw sequence data for each sample were subjected to de novo assembly using GetOrganelle 1.7.7.0 (Jin et al. 2020). The software GeSeq from the toolkit Chlorobox (Tillich et al. 2017) was used to annotate genes in the resulting chloroplast genome assemblies. The sequences of the aforementioned highly conserved genes, along with 1000-bp flanking regions on either side of the gene, were extracted from their respective annotation files to include in alignments with the data obtained from GenBank.

The de novo assemblies resulted in circularized genomes for 14 samples, with base coverages ranging from 67.5X to 1509.8X and an average chloroplast genome length of 153,875 bp (Table S1). Uncircularized or

partial assemblies were obtained for 17 samples, while two samples were completely unsuccessful in generating any assembled sequences. For 28 samples at least one of the four target chloroplast markers (*rbcL*, *matK*, *ndhF* and *trnF*) was assembled and annotated. For 23 samples, all four markers were annotated and included in the analysis.

Sequence Alignment and Phylogeny—An initial sequence alignment for each marker was created using MAFFT version 7 (Katoh and Standley 2014). The software Geneious Prime version 2024.0.5 was used to manually inspect the sequence alignments, editing the sequences to remove long strings of entirely ambiguous bases (encoded as N), as well as to trim sequence ends and to refine the alignments. In some cases, some low-quality sequences were removed entirely from the alignment. The alignments were refined using the ClustalW algorithm (Sofi et al. 2022) within Geneious Prime, with gap penalties set to default values. After alignment, low-quality ends and ambiguous regions were trimmed from each sequence to retain only the high-quality, core region of the multiple sequence alignment. To identify potential issues with alignment and misidentified sequences, a single-marker phylogeny was generated for each marker using RAXML v8.2.11 (Kozlov et al. 2019), and these issues were resolved for each of the alignments prior to concatenation of the 5 single-marker multiple sequence alignments using a custom Python script. The final multi-marker multiple sequence alignment was 13,217 bp in length and contained 1696 sequences, including the newly generated sequences from 31 herbarium specimens (Hossain et al. 2025). This alignment was used to infer a final maximum-likelihood phylogenetic tree with RAXML, using a GTRGAMMA nucleotide substitution model and five data partitions. The resulting phylogenetic tree was visualized with iTOL (Letunic and Bork 2024). The phylogenetic tree comprises 7 classes, 67 orders, 231 families, and 1696 genera. The placement of each branch tip was checked for consistency with the reference phylogeny from Kew Tree of Life Explorer (Baker et al. 2022) and confirmed to belong to a monophyletic group at the order level. In visualizing and rooting the tree, the class Lycopodiopsida was selected as an outgroup based on an available phylogeny of vascular plants (Cole et al. 2019).

Spatial Phylogenetic Analysis—A spatial phylogenetic analysis was performed using the R package “canaper” (Nitta et al. 2023). Initially, the community matrix data was imported for the 15 × 15 km area (0.134747 degrees) and a phylogenetic tree using the “ape” package in R (Paradis et al. 2004). The community matrix was obtained from resolution and redundancy analysis for the spatial phylogenetic analysis with redundancy values greater than 0.1. The R package picante (Kembel et al. 2010) was used to match the genera from the community matrix data with the tip labels of the phylogenetic tree. Since the phylogeny was constructed with 1696 taxa, the community data matrix does not contain all the same taxa due to cleaning and completeness filtering of the occurrence data through redundancy analysis. The “match.phylo.comm” function aligns the phylogenetic tree and the community matrix, ensuring that the data matched for further analyses.

In spatial phylogenetic analysis, randomization is an essential process used to assess whether the observed patterns of PD are statistically different from what is expected by chance under a null model. The statistical outcome is largely dependent on the number of randomizations. To identify the appropriate randomization number, at first the “cpr_iter_sim” and “n_reps_test” functions under the “curveball” null model were performed, which allowed the determination of the number of iterations and number of replicates. These are two different values defined as *n_reps* for a number of replicates as random grid-cells to simulate during randomization, and *n_iterations* refers to the number of times a randomization or simulation is repeated. A minimum of 1000 replicates was chosen with 10,000 iterations under the “curveball” null model. This “curveball” model shuffles species occurrences in each grid cell while preserving species richness in each grid cell and frequency of occurrence of each species (Strona et al. 2014). Therefore, the “cpr_rand_test” function from the Canaper package was used to obtain the randomization which returns the PD, PE, RPE, and RPD. Finally, the “cpr_classify_endem” function was used to classify the results into types of endemism (neo- or paleo-endemism) based on the significance of the deviations in PD.

In the CANAPE analysis, first a one-tailed test was performed with the PE original tree and the PE comparison tree to find the centers of endemism. Next, a two-tailed test was performed with RPE, where the regions that were significant in the one-tailed test were identified. Regions showing significantly high PD relative to random expectations indicate phylogenetic overdispersion, while those with significantly low PD suggest phylogenetic clustering. RPD and RPE are calculated using the ratio of the original tree of PD and PE with the comparison tree. The comparison tree has the same nodes, but is reduced to a cladogram in which all branches

are of equal length. The original tree of PD and PE is considered as the numerator, and the comparison tree is considered as the denominator. Since the cladogram treats all divergence events as equal length, any deviations from the original tree describe the evolutionary history of lineages. If the ratio is larger, it indicates the presence of longer branch lengths, suggesting older lineages. On the other hand, a smaller ratio indicates the presence of shorter branch lengths, which suggests recent diversification.

Elevation data were downloaded to identify relationships with cluster groups and CANAPE metrics. Elevation data were obtained from the WorldClim version 2.1 database using the “worldclim_global()” function in the R geodata package (Hijmans et al. 2025). We used boxplots to visualise the elevational ranges of different CANAPE endemism categories, cluster groups as well as cells with significant PD and RPD (Fig. S5). Country boundaries were obtained from Natural Earth (Massicotte and South 2025), and additional administrative boundaries were obtained from the GADM database of Global Administrative Areas (Hijmans et al. 2025). All spatial data processing was conducted using the R package “sf” (Pebesma 2018), and “terra” (Hijmans 2024). All the plots in this study were made by using R package “ggplot2” (Wickham 2016).

Phylogenetic Turnover and Floristic Composition—The phylogenetic turnover analysis was calculated using the “phylobeta” function from the “phyloregion” R package, which assesses phylogenetic beta diversity (Daru et al. 2020). The Sørensen index was chosen to measure the dissimilarity between communities based on shared phylogenetic branch lengths. To assess the phylogenetic turnover among communities, the community data matrix and phylogenetic tree were employed in the phylogenetic turnover calculation. Ward’s hierarchical clustering method (Ward.D) (Murtagh and Legendre 2014) was performed on the turnover dissimilarity matrix to identify groups of communities with similar phylogenetic composition. The hierarchical clustering object was converted into a phylogenetic tree format using the ape package function (as.phylo) (Paradis et al. 2004) and visualized as the dendrogram with “ggtree” (Yu et al. 2017).

The optimal number of clusters was determined using the NbClust function (Charrad et al. 2014), which is a consensus-based approach that runs different clustering algorithms multiple times and results in robust estimates of optimal cluster numbers. The NbClust function was applied on principal coordinates analysis (PCoA) reduced data, based on the phylogenetic turnover distance matrix. The majority of the indices supported $k = 4$ as the optimal number of clusters (Fig. S4), and were chosen for further analysis. Selection of the optimal number of clusters (k) poses complex challenges, as clustering algorithms will always find a k value even when no true cluster exists. Relying solely on a particular method would be a risk of misinterpretation. Therefore, the suggested k value should be interpreted with reference to the ecological and biogeographical context.

To identify the floristic composition of each phylogenetic turnover cluster group, we conducted an indicator species analysis. The multipatt function was applied with the group-equalized indicator value (IndVal.g) index, as described in the “indicspecies” R package (De Cáceres and Legendre 2009). We used a genus-by-grid-cell abundance matrix, with absences coded as 0 and the cluster label for each grid cell. IndVal.g combines two components: specificity, defined as the degree to which a genus is restricted to a particular cluster, and fidelity, representing the proportion of sites in the target cluster where the genus is present. The evaluated indicator value index (IndVal.g) measures each genus’s association with each cluster. To test the statistical significance of associations, we performed 999 permutations. Genera with $p \leq 0.05$ were retained as significant indicators of the corresponding cluster.

RESULTS

Spatial and Molecular Data—The resolution and redundancy analysis plot showed that while finer grid cells (5–10 km) suffer from insufficient sampling, larger sizes (20–25 km) provide better coverage but at the cost of losing spatial details (Fig. S1). This emphasizes the need for a balanced approach. The 15×15 km grid size achieves this balance, ensuring fewer poorly sampled areas and a reasonable number of adequate sampled cells. The GBIF occurrences were distributed in 15×15 km resolution with 422 grid cells projected in Fig. S2b. The removal of low-occurrence grid cells based on the redundancy and resolution value, and further matching the phylogenetic

tree with community data matrix, eliminated 61 grid cells resulting in 361 grid cells for final analysis (Fig. S2a). This dataset was subsequently used to explore all the species diversity metrics and phylogenetic diversity analysis. The selected 567 genera account for a total of 119,909 occurrence records selected for analyzing all diversity metrics. The phylogeny was constructed using the initial taxonomic selection, prior to the removal of flagged coordinate data comprising 1696 taxa. The constructed phylogenetic tree is displayed with monophyletic order labels (Fig. S6). In the phylogeny, Polypodiales exhibited the longest branch lengths. In addition to Polypodiales, the genera *Vismia* (Hypericaceae, Malpighiales) and *Cuscuta* (Convolvulaceae, Solanales) also exhibited relatively long branch lengths, as has been observed in previous studies from both Chile and Norway (Scherson et al. 2017; Mienna 2018).

Diversity Patterns—The diversity patterns of taxon richness (TR) and observed phylogenetic diversity (PD) (Fig. 2A, B) were highly correlated as expected (Fig. S3). The diversity gradient was not evenly distributed across the region, but followed a central-high, north-and-south-low pattern within the ecoregion. There is a notable decline as a central-to-peripheral gradient for both TR and PD values. Higher diversity patterns were observed in approximately 15–20% of the cells in the Ecuadorian cordillera and central cordillera in Colombia. On the other hand, the lowest diversity gradient was observed in approximately 30% of the cells in Venezuela and the southernmost part of Peru. The occurrence sampling (Fig. S2b) likely influences the TR and PD gradient, as areas with higher numbers of occurrences show higher diversity indices, which could affect the observed gradient.

The significance of PD and RPD across the páramo ecoregion are presented in Fig. 2C and Fig. 2D. The distribution indicates that no specific region had notable clusters of significantly high or low values; most are distinct and scattered throughout the region. Several regions with significantly low PD were observed throughout the region, except in the Venezuelan páramo. Conversely, highly significant PD was rare, with only a few regions exhibiting high PD, observed only in the Venezuelan and Ecuadorian sections of the páramo. The observed diversity pattern of RPD significance demonstrates a notable concentration of high RPD in Venezuela and in the Ecuadorian Cordillera, with additional isolated pockets of high RPD regions scattered across the Colombian páramo region. The low RPD is observed throughout the region as distributed in the isolated pockets and a notable concentration in the Colombian Central Cordillera. The overall trend was that lower PD values were predominant over higher values and RPD significance was infrequent, appearing in sparse and isolated cells across the páramo region.

Endemism Patterns—The observed distribution of weighted endemism (WE) and phylogenetic endemism (PE) mirrored their spatial patterns throughout the region (Fig. 3A, B). Both indices indicated the highest concentrations in Ecuador and Colombia, with lower values observed in northern Venezuela and southern Peru. Generally, the spatial patterns of WE and PE closely correspond across the region, although the intensity varies because of the differences in WE and PE values.

The PE and RPE significance distributions were very distinct in their diversity patterns (Fig. 3C, D). For PE, cells with significantly low values were most prevalent and tended to show a widespread distribution, including noticeable clusters throughout the Eastern Cordillera, the Ecuadorian Cordillera,

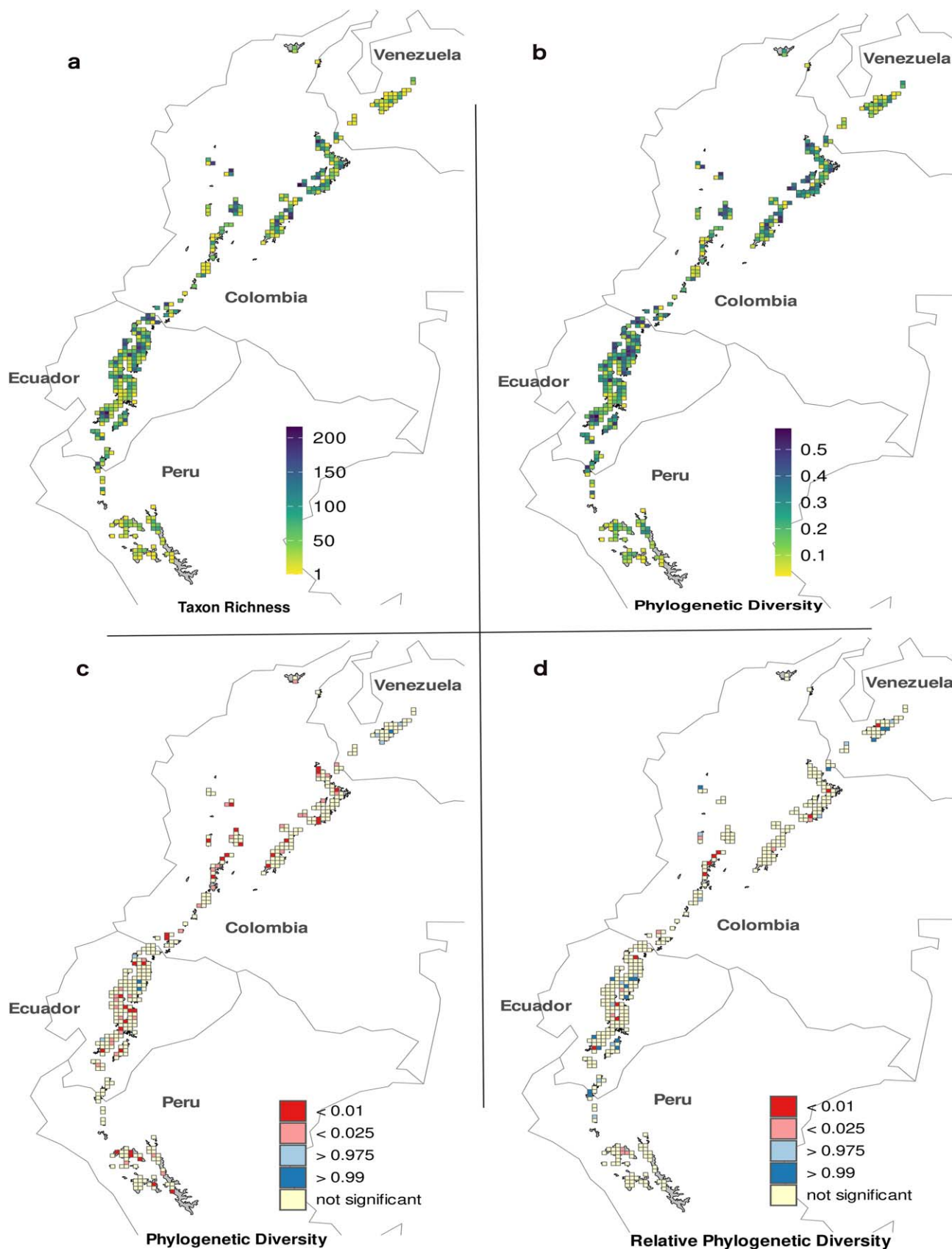


FIG. 2. Spatial distribution of diversity patterns across the páramo region. A. Taxon richness (TR) represents the total number of genera found in each cell. B. Phylogenetic diversity (PD) indicates the sum of branch lengths for the genera present in each cell. C. Randomized PD indicates areas with significantly high or low phylogenetic diversity. D. Randomized relative phylogenetic diversity (RPD) identifies areas with high concentrations of long branches and short branches. For C and D, significance levels are indicated as follows: blue cells show significantly high values (> 0.975), red cells indicate significantly low values (< 0.025), and yellow cells denote regions where values are not statistically significant.

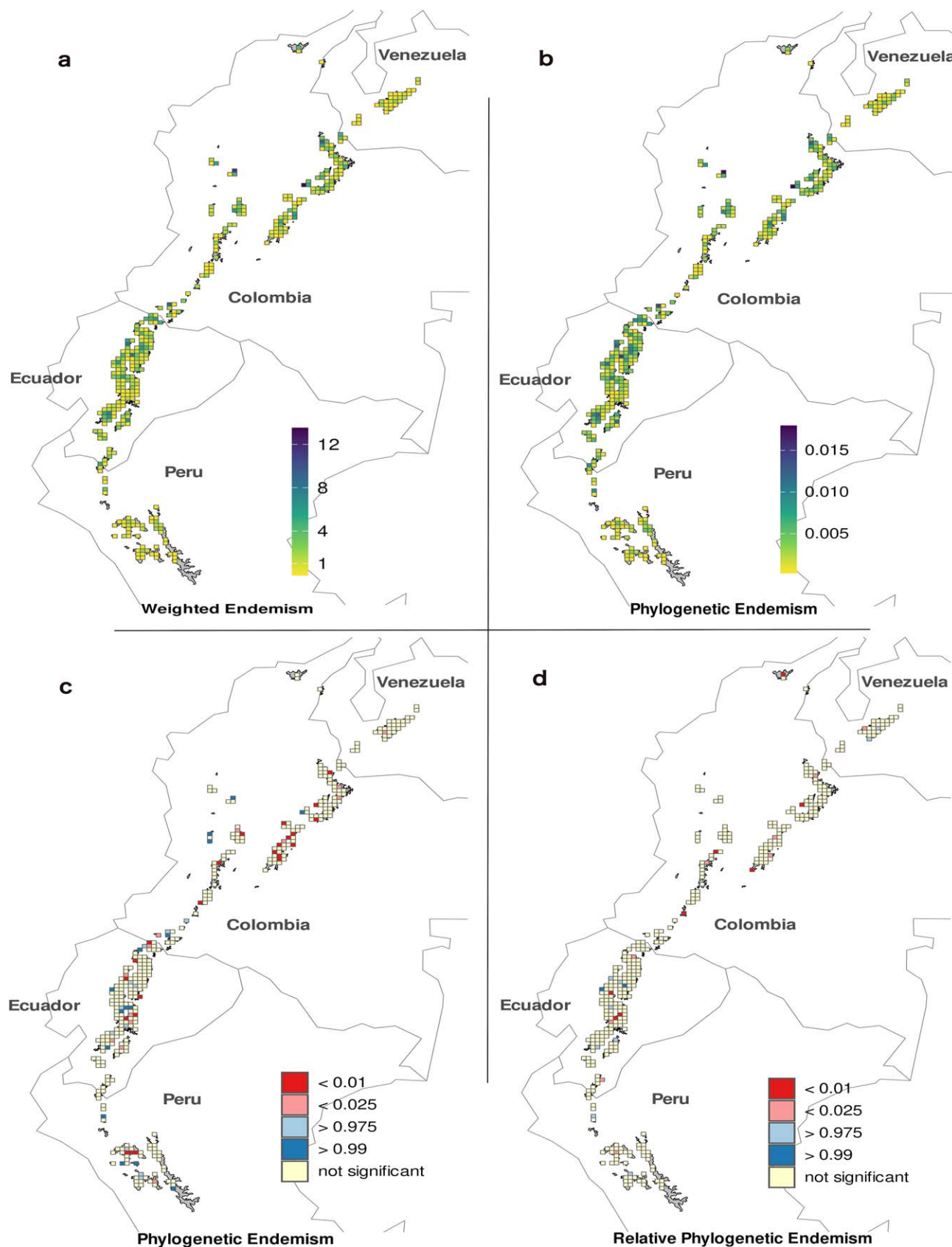


FIG. 3. Spatial distribution of vascular plant endemism across the Northern Andean páramo. A. Weighted endemism calculated as the sum of branch lengths in each cell multiplied by each branch's restriction proportion. B. Phylogenetic endemism calculated as the number of taxa present in each cell multiplied by each taxon's restriction proportion. C. Phylogenetic endemism significance. Highlights regions with significant concentrations of phylogenetic endemism (unique evolutionary lineages). D. Relative phylogenetic endemism significance. Identifies areas with significant concentrations of phylogenetically restricted lineages, where high values (blue) indicate regions with a concentration of long branches and low values (red) indicate high concentrations of short branches.

and some isolated cells in Peru. In contrast, cells that showed significantly high PE values were much less frequent, primarily located in Ecuadorian Cordillera and in scattered regions in Colombia. The RPE distribution, however, showed an entirely different pattern, with little correspondence to the PE distribution. The overall number of significant RPE cells was very low, and there was no clustering; these cells were sparsely distributed across the region.

The CANAPE results identified different types of endemism across the páramo region without showing strong spatial clustering (Fig. 4). Apart from the non-significant cells, mixed-endemism was the most prevalent category. It was widely distributed with 49 cells throughout all páramo ecoregions. The second most common category found was paleo-endemism, observed in 9 cells, mostly scattered across Peru, Ecuador, and Venezuela and absent from the Colombian páramo. A total of 7 cells exhibit super-endemism scattered throughout the region. Only 5 cells showed neo-endemism which were isolated and scattered, without forming any spatial clusters.

Turnover Patterns—Clusters 1 to 4 showed distinct geographic distribution patterns across the páramo regions (Fig. 5). Cluster 1 (purple) was the largest, encompassing nearly half of the total sites and spanning a wide geographical area across the páramo region. This group was dominant in the Venezuela and northern Colombia belt, specifically in the Cordillera de Mérida and the Eastern Cordillera. It continues southward as patchier occurrences through Ecuador and into Peru. This cluster was spatially coherent, with minimal fragmentation, and was present across most páramo regions. Cluster 2 (yellow) accounted for nearly 19% of the total sites. Its distribution was moderately continuous throughout the Ecuadorian Cordillera and into northern Peru. However, as it extends northward into Colombia, Cluster 2 becomes patchier, occurring as scattered occurrences interleaved with other clusters. Compared with Cluster 1, Cluster 2 is less widespread overall but well represented across multiple páramo subregions, with a particular concentration in mid-to-southern areas. Cluster 3 (green) was primarily continuous along the Ecuadorian Cordillera, extending into northern Peru. Outside of this core area, it appears only sporadically to the north, with only a few isolated cells found in Venezuela. Cluster 4 (red) displayed a scattered and patchy distribution in the Colombian Western Cordillera, becoming increasingly sparse as it extended southward into the Ecuadorian Cordillera. This cluster was geographically isolated and completely absent from the Peruvian páramo. The distributions of mean elevation for CANAPE endemism categories, cluster groups, PD significance, and RPD significance (Fig. S5) showed largely overlapping ranges.

Floristic Composition—Indicator species analysis identified 176 genera significantly associated with four phylogenetic turnover clusters. Cluster 4 contained the highest number of indicator genera, with 87 assigned to it. Clusters 2 and 3 comprised 37 and 42 genera, respectively, while Cluster 1 included only 10 genera. Cluster-specific plots (Tables S3–S6) showed the distribution of indicator genera by indicator value. In Cluster 1, *Espeletia*, *Brachyotum*, *Escallonia*, and *Hesperomeles* are the primary contributors. Cluster 2's primary indicator genera were *Carex*, *Agrostis*, *Cerastium*, and *Gentianella*. Cluster 3 is dominated by *Solanum*, *Calceolaria*, *Bidens*, and *Muehlenbeckia*. Cluster 4 was the most diverse, comprising genera such as *Anthurium*, *Palicourea*, *Miconia*, and *Begonia*.

DISCUSSION

This study used spatial phylogenetic analysis to explore vascular plant diversity in the páramo ecosystem from the perspectives of both evolution and endemism. The analysis identified specific geographic areas that harbor ancient lineages and recently diverged taxa, and those zones containing the highest and lowest phylogenetic diversity. A substantial pattern of phylogenetic clustering across most regions indicates strong abiotic filtering and niche conservatism that favors closely related taxa adapted to the páramo's unique and harsh high-altitude conditions. This study also identified several distinct regional hotspots characterized by mixed endemism, suggesting that the region harbors both ancient relictual lineages that preserve deep evolutionary history and recently diverged lineages that contribute to the region's rapid diversification. The phylogenetic turnover analysis reflects the evolutionary structure of the páramo flora by revealing both widespread shared histories and localized, distinct assemblages shaped by environmental and geohistorical factors, effectively delineating in distinct bioregions across the páramo.

The analyses revealed that areas with more extensive sampling tend to exhibit higher taxon richness and observed phylogenetic diversity, suggesting possible sampling bias, as noted in a study of Chilean flora (Scherson et al. 2017). In remote areas like the páramo ecoregion, uneven sampling and data gaps may occur due to geographic barriers and varying observer efforts (Hortal et al. 2007). By employing randomization and comprehensive statistical controls, it is possible to find meaningful diversity patterns in the region (Scherson et al. 2017). Another methodological approach to mitigate this bias is the implementation of rigorous spatial data cleaning and the standardization of all taxonomic names according to a unified taxonomy, both of which were applied in this study. Additionally, selecting the optimal grid cell size based on sampling completeness and redundancy values enhances data quality for subsequent analyses, as was also performed in this research.

The high correlation between TR and observed PD was expected (Faith 1992; Forest et al. 2007), and similar levels of association have been noted in previous studies (Scherson et al. 2017; Mienna 2018), given that higher taxon richness corresponds with greater evolutionary lineage diversity. The overall trends of taxon richness across páramos are largely consistent with existing literature (Rangel-Churio and Pinto-Zárate 2012; Cuesta et al. 2017; Peyre et al. 2019; González-Orozco and Parra-Quijano 2023). However, direct comparisons with previous research are difficult due to differences in geographic scale and regional delineation (Peyre et al. 2015). Previous studies have confirmed high diversity in Ecuadorian and Colombian páramos (Cuesta et al. 2017). The lowest floristic diversity in Venezuela has also been documented (Peyre et al. 2015). The present study demonstrates a decrease in taxon richness from central to peripheral regions, specifically the north and south. This pattern likely results from the inclusion of all major páramo regions in the analysis, rather than examining them individually. The high PD in the region indicates higher genetic diversity and longer branch lengths within taxa. Lower PD suggests less phylogenetically diverse and more recent common ancestry (Faith 1992). Both the TR and PD indices show lower diversity in Venezuelan and Peruvian páramo. The presence of the Huancabamba

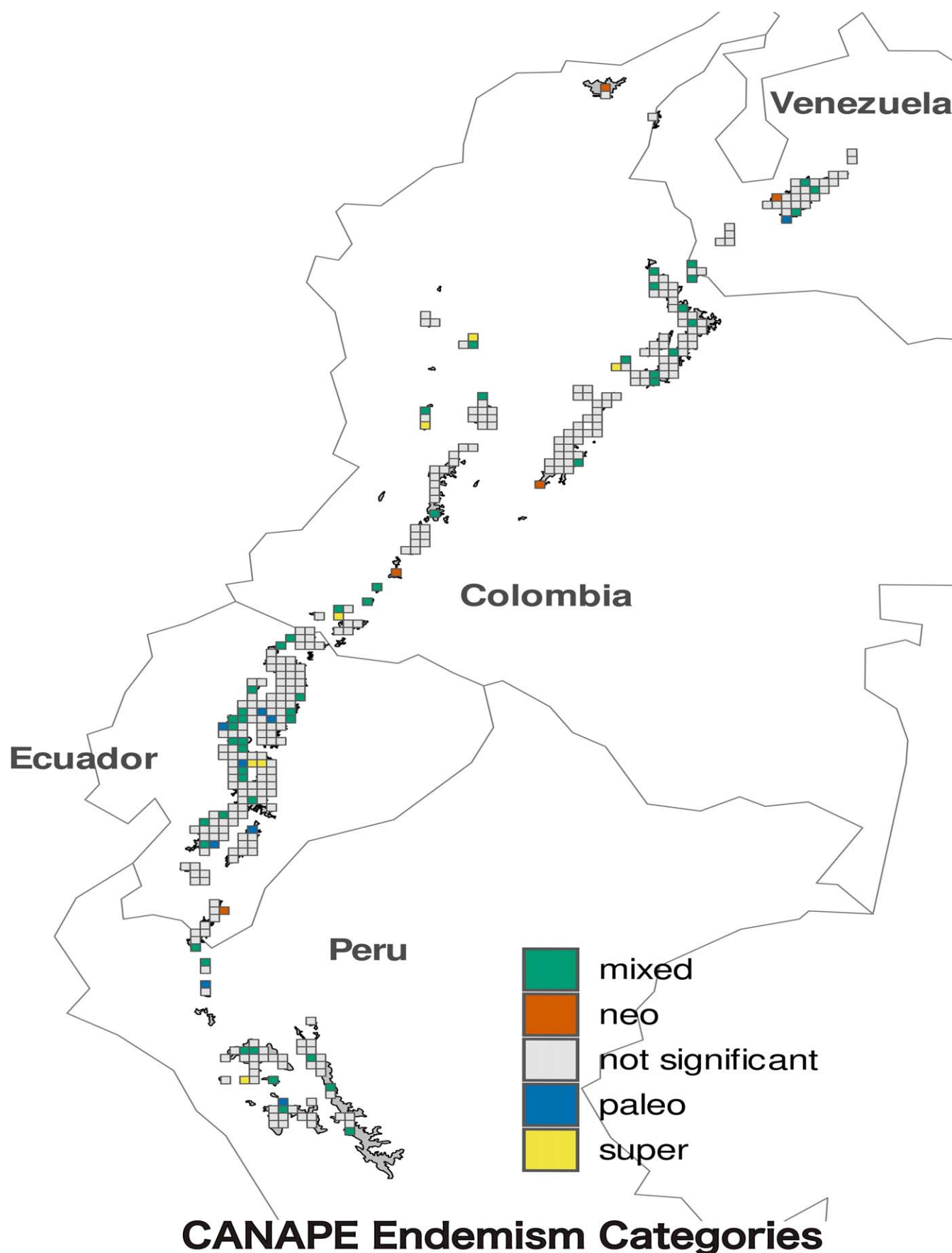


FIG. 4. Distribution of endemism categories across the Northern Andean páramo. Cells are color-coded to represent different endemism types: red cells (neo) indicate concentrations of significantly short restricted branches, representing neo-endemism; blue cells (paleo) indicate concentrations of significantly long restricted branches, representing paleo-endemism. Green cells (mixed) show areas with concentrations of both short and long branches. Yellow cells (super) represent areas with exceptionally high endemism, while gray cells are not statistically significant.

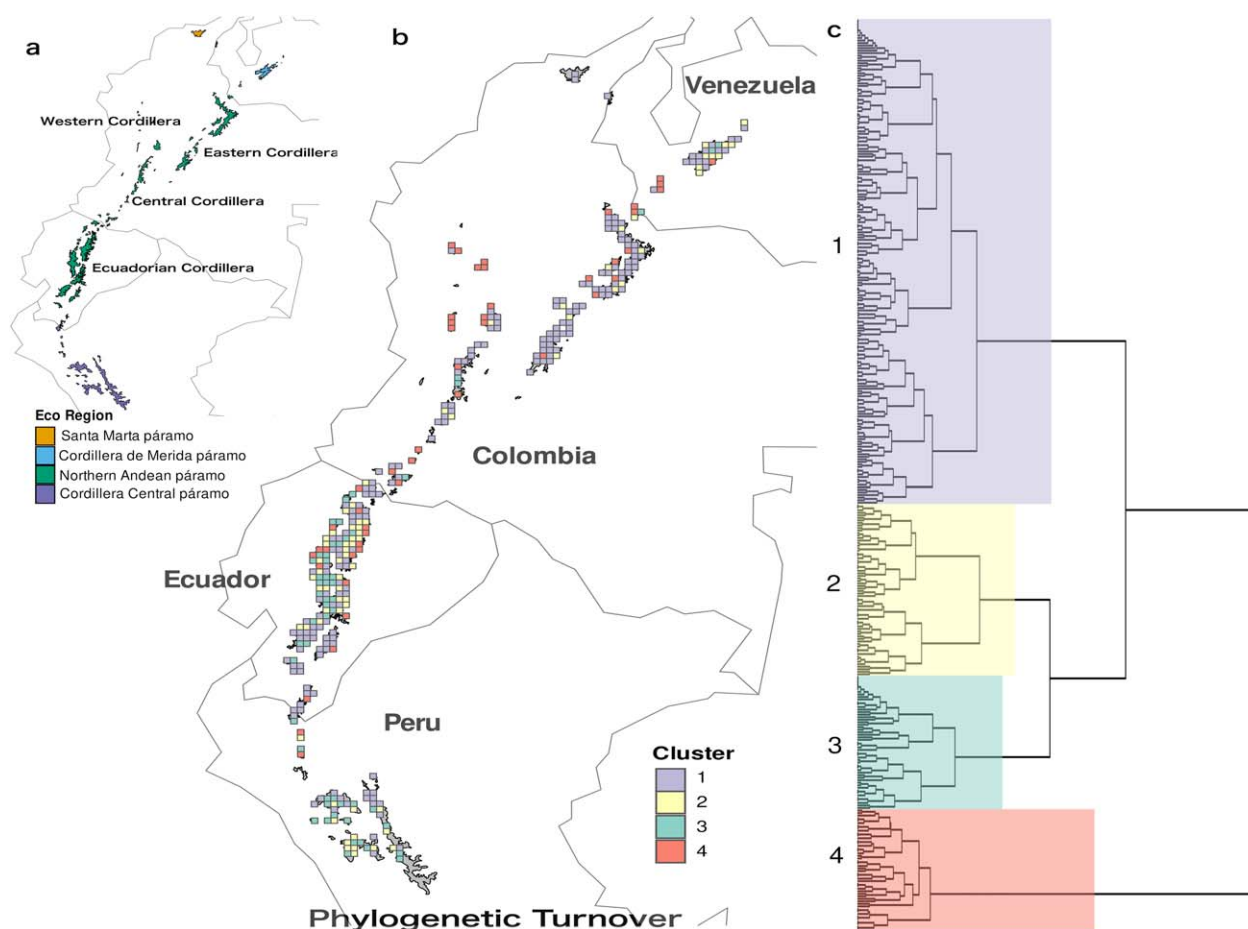


FIG. 5. Hierarchical clustering and distribution of páramo plant communities according to the phylogenetic turnover analysis. The phylogenetic turnover matrix resulted in four distinct groups of plant communities. A. Inset map of Andean páramo ecoregion. B. The map illustrates the clusters, with colors representing different groups. It compares all grid cells and groups them based on the phylogenetic branches they share. C. The turnover topology displays the hierarchical relationships among species assemblages, where each clade is color-coded to match the cluster groups shown on the map.

Depression, a low-altitude biogeographical barrier (~2145 m), is one possible explanation for the lower diversity in Peruvian páramos, as it could disrupt the continuity of montane ecosystems and act as a dispersal barrier for highland taxa (Weigend 2002; Quintana et al. 2017).

The pattern of PD significance indicates significantly low PD cells present in almost every páramo region except in Venezuela (Fig. 2C), indicating phylogenetic clustering where taxa are more closely related than expected by chance. This may result from abiotic filtering (Webb et al. 2002; Swenson et al. 2007) through which the differential tolerance of taxa to specific non-living environmental factors determine which taxa can be present. Similarly, closely related taxa may coexist in environmentally challenging habitats because their similarities might reduce competition (Callaway et al. 2002; Bruun et al. 2006). In harsh or stressful environments, such as cold and dry regions, only particular clades may have evolved the necessary adaptations to survive, leading to phylogenetic clustering (Spalink et al. 2018). The unique and challenging high-altitude climatic conditions of the páramo support this interpretation, which is consistent with findings from studies in similarly demanding environments, such as the colder, drier regions of northern Norway (Mienna et al. 2020) or the arid interior of Australia (Thornhill et al. 2016). Abiotic filtering prevents taxa from extending their range and colonizing these regions and might be associated with páramo's unique

environmental conditions (Luteyn 1999) and past temperature fluctuations during glacial oscillation (Madrinán et al. 2013). Surviving taxa often cluster phylogenetically, reflecting niche conservatism, where closely related species maintain ancestral traits and occupy similar niches (Losos 2008).

Significantly high PD values, which indicate phylogenetic overdispersion (Webb et al. 2002; Swenson et al. 2007) were rare and observed in a few Venezuelan and Ecuadorian páramo cells. Phylogenetic overdispersion suggests that taxa in these regions are less closely related than expected by chance (Fig. 2C) (Mishler 2023), indicating that biotic interactions may favour coexistence among distantly related taxa. Such patterns are often observed in environmentally favorable conditions (Mienna et al. 2020) where competition among close relatives promotes niche partitioning and resource differentiation (Swenson et al. 2007). However, interpretation of overdispersion is complex and depends on the interplay between niche differences, which promote coexistence, and differences in competitive ability, which can drive exclusion. This balance between differentiation and competitive ability allows distantly related taxa to coexist within a community (Mayfield and Levine 2010; Kraft et al. 2015).

Significantly high RPD values were infrequent and appeared in sparse, isolated cells across the páramo region (Fig. 2D). However, there was a noticeable concentration of

high RPD values in Venezuelan páramo and Ecuadorian cordilleras, with additional isolated pockets of high RPD regions scattered across the Colombian páramo. Cells with significantly high RPD values indicate areas with long branches that may be home to biogeographic relict taxa. Some of these cells are also characterized by high PD, suggesting they contain older evolutionary lineages that have persisted through past ecological shifts and extinction events (Thornhill et al. 2016). In contrast, significantly low RPD values were mostly found in the Colombian Central Cordillera and in the Ecuadorian Cordillera, indicating regions of recent diversification (Thornhill et al. 2016).

The biogeographic history of the páramo can be interpreted through the lens of its rapid diversification, complex origins, and fragmented “sky island” nature. The páramo’s biogeographic history, niche shifts, and range expansions along the Andean cordilleras have influenced the diversification of high-elevation endemic plants (Kolár et al. 2016). Notably, the Venezuelan páramo was characterized by the presence of highly significant PD and RPD, which supports the coexistence of distantly related plant lineages in the region. Additionally, lineages present in the Venezuelan páramo were often ancient, which supports that this region may be a refugium for relict plant taxa, as reported in a previous study on the Asteraceae family (Diazgranados and Barber 2017). The presence of high PD and RPD here suggests these areas have retained a deeper evolutionary history, possibly due to their specific uplift history, sustained environmental stability, or unique colonization events. The low PD value in the Eastern Cordillera of Colombia suggests strong environmental regulation acted as a filter, allowing only certain, closely related plant lineages to establish. Similar trends were observed in the Colombian Central Cordillera, where additional low RPD suggested recent diversification within certain, closely related plant lineages. Páramo flora experienced rapid diversification and multiple radiations during the Pleistocene, and the net diversification rate in the páramo is the fastest among all biodiversity hotspots (Madriñán et al. 2013). This rapid diversification can be attributed to ecological opportunities that arose following the uplift of the Andes during the late Pliocene and Pleistocene, which led to the formation of new habitats and niches that facilitated speciation (Hughes and Eastwood 2006). The Ecuadorian Cordillera presents a complex and intriguing evolutionary history, characterized by both strong environmental filtering (low PD) in the southern regions and the persistence of ancient lineages (high RPD) in the northern parts of the region. The Peruvian páramo is considered to comprise a closely related taxa existing together with low PD. This suggests that climatic fluctuations may have influenced speciation, leading to population isolation and habitat fragmentation that facilitated the diversification process (Anthelme et al. 2014; Flantua et al. 2019). The overall observed diversity patterns also corroborate previous conclusions that geological events, such as Andean uplift and past climatic fluctuations during the Miocene and Pleistocene, as well as the profound influence of the páramo’s fragmented geography, have had on the diversity patterns.

The CANAPE analysis identified and categorized centers of endemism across the páramo ecoregion. The most commonly identified endemism categories were mixed and super-endemism. Mixed-endemism cells suggest the region hosts both recently diversified taxa and ancient lineages. This reflects a combination of recent speciation and the presence

of long-branched taxa. The largest aggregation of mixed-endemic cells was found in Ecuador, but these cells were also distributed throughout the entire páramo region. Paleo-endemic cells were harboring ancient lineages that were once widespread but are now confined to specific locations. These putative evolutionary refugia occurred in several isolated patches throughout the entire ecoregion. Super-endemic cells indicate regions with many taxa that are extremely range-restricted and have long-branched distributions. These cells were also primarily scattered across the páramo region. Neo-endemic cells were found scattered in a few regions and occurred in isolated patches. These indicate regions with taxa that have evolved only recently and remain endemic due to restricted dispersal or migration.

Each of the four identified clusters of páramo plant communities showed a distinct geographic distribution pattern across the páramo ecoregion, reflecting spatial patterns in phylogenetic composition. The continuity of Cluster 1 across most páramo regions, combined with its extension southward into Ecuador and Peru, highlights phylogenetic similarity widely shared across multiple páramo subregions. This suggests that Cluster 1 represents a foundational floristic assemblage, underscoring its importance as a broad biogeographic unit. The distribution of Cluster 1 in Venezuela and Colombia aligns with the findings of Londoño et al. (2014), which identified the Eastern Cordillera as the largest and most cohesive floristic province. This area encompasses Mérida in Venezuela, recognized as a distinct phytogeographic region with high levels of endemism. Similar to Cluster 1, Cluster 2 was also well-represented across multiple páramo subregions, with most concentrations found in the Ecuadorian Cordillera and other southward-leaning regions. The wide-spread distribution of Clusters 1 and 2 provides phylogenetic evidence that floristic similarities are also widespread and exist beyond national borders. Cluster 3’s primary concentration in the Ecuadorian Cordillera, with extensions into Peru, suggests it as a distinct southern páramo assemblage and also indicates the existence of regional indicator lineages or strong habitat filtering unique to this cordillera. Cluster 4 showed the most geographically isolated and patchy distribution with a more northerly distribution. Its absence in Peru corresponds with the Huancabamba Depression, a well-documented biogeographic barrier that limits the dispersal of many high-Andean taxa (Kolár et al. 2016). The distinct pattern observed is that Clusters 1 and 4 are mostly dominant in the northern regions, while Clusters 2 and 3 are more abundant in the southern regions (e.g., the Ecuadorian Cordillera and Peru).

The indicator genera identified within the cluster group include floristic elements representing several origins of páramo flora. As described by Sklenář et al. (2011), all phylogenetic turnover clusters show biogeographic affiliations corresponding to distinct floristic origins. However, the significance of the indicator genera varied across clusters. For example, Cluster 1 is characterized by a combination of páramo endemics, with notable representation of Neotropical flora and elements from the Austral–Antarctic region. Cluster 2 was dominated by wide-temperate genera but also contains Holarctic immigrants and some Austral–Antarctic genera. Cluster 3 was largely shaped by Neotropical elements, as well as Holarctic and Austral–Antarctic elements, and included some cosmopolitan genera. Similarly, Cluster 4 was strongly defined by Neotropical taxa, including páramo

endemics and wide tropical elements. The distinct phylogenetic assemblages are likely the result of past biogeographic processes, including historical connectivity and long-distance dispersal events, which were facilitated by niche shifts and range expansions along the cordilleras (Kolář et al. 2016). Overall, the páramo's phylogenetic composition was heterogeneously distributed, reflecting the region's complex history, local diversification, geographic barriers, and ecological filtering.

For the first time, this study generated and analyzed a genus-level phylogeny encompassing the entire vascular flora of the páramo, which included novel genetic sequences from the chloroplast genomes of 31 páramo plant taxa. The spatial phylogenetic approach revealed the phylogenetic diversity and endemism patterns of the vascular flora in the Andean páramo. The findings revealed distinct regions of high endemism, including neo- and paleo-endemism areas, emphasizing the importance of conserving these regions containing unique evolutionary lineages. Given the páramo's numerous threats, these findings emphasize the importance of targeted conservation in those areas based on this evolutionary perspective. While conservation decisions and funding are often limited by political boundaries, which leads to overlooking the patterns of local diversity and endemism (Mishler et al. 2020), conservation efforts should prioritize areas of local endemism that contain high concentrations of rare taxa (Thornhill et al. 2017). In this regard, with the páramo's highlighted endemism distributed across four countries in several patchy and isolated regions, these regions should therefore be prioritized by local conservationists within the framework of regional-scale planning. The most common endemism category, mixed-endemism, is highly important to conservation as it indicates both recent diversification and concentrations of relictual taxa. Protecting these areas would conserve ecosystems that facilitate ongoing evolutionary processes while also preserving refugia for older, range-restricted taxa. Future studies should overlay the mixed-endemism and high-endemism regions with existing protected area networks to measure the effectiveness of the existing protected area network and suggest conservation approaches based on the findings from this study. Future conservation plans for the páramo should integrate spatial phylogenetic findings with climate change projections to identify which endemism centers are most vulnerable to predicted shifts. Further investigation of the abiotic factors (e.g., temperature, precipitation, topography, soil type) influencing the observed metrics would reveal the bio-climatic drivers of phylogenetic diversity and endemism in the páramo ecoregion.

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AUTHOR CONTRIBUTIONS

JDMS, MDM, and AZ conceived the project. SLFM, KH, and MDM performed the laboratory work. RDLT performed the chloroplast genome assemblies and extracted the relevant marker sequences. KH performed the informatic and phylogenetic analyses. KH wrote the paper with contributions from all authors. AZ provided botanical expertise and logistical support for the herbarium sampling. JDMS, MDM, SLFM, and AZ supervised the research. The research was supported by the Norwegian Directorate for Higher Education and Skills via NORPART award 2021/10475 ("BiGTREE: Biodiversity Genomics Teaching, Research, Exchanges and Education").

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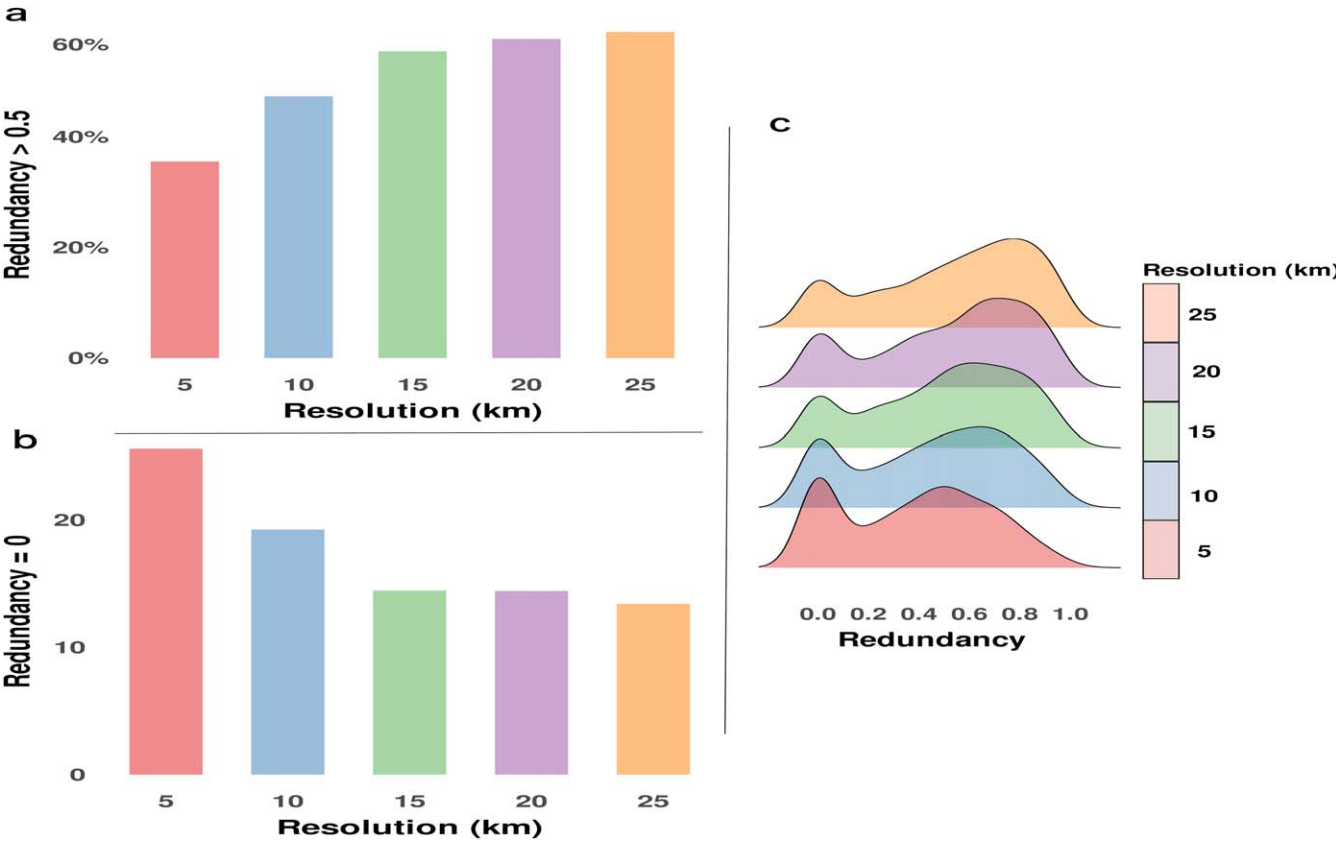


FIG. S1. Dependence of sampling redundancy and resolution on the choice of grid size. A. Percentage of grid cells with redundancy > 0.5 at different grid sizes (5 km, 10 km, 15 km, 20 km, and 25 km) shows that larger grid cells (20–25 km) tend to have better sampling coverage but reduced spatial detail. B. Percentage of grid cells with redundancy = 0 indicates a reduction in poorly sampled areas with larger grid cells. C. Distribution of redundancy values by grid size with a balance at 15-km resolution provides a balance between fewer poorly sampled areas and maintaining spatial detail.

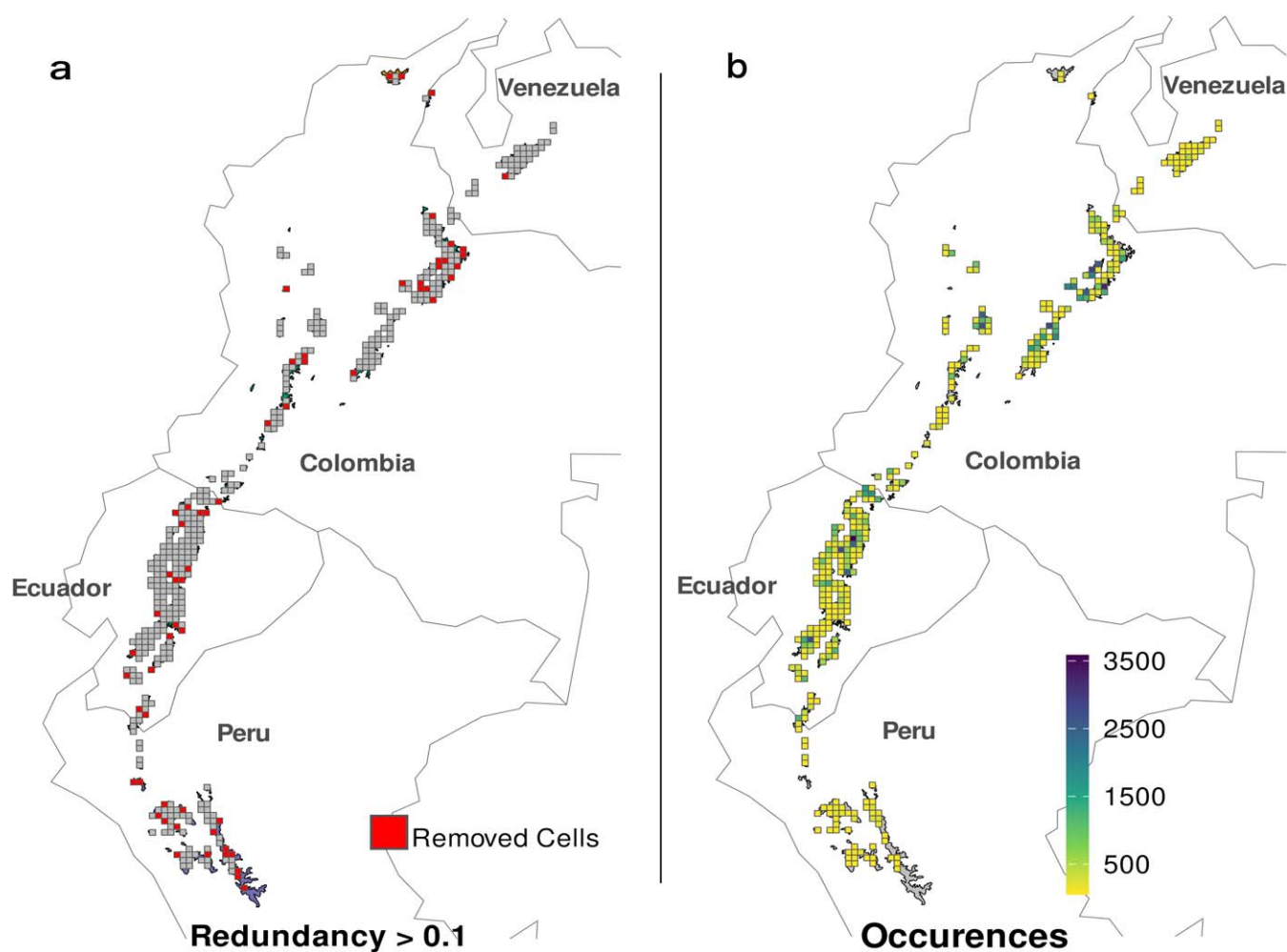


FIG. S2. Redundancy and occurrence distribution across the Northern Andean páramo. A. Map displaying cells with redundancy values below 0.1, marked in red, which were excluded from the analysis. B. Occurrence map illustrating the distribution of species occurrences across the páramo.

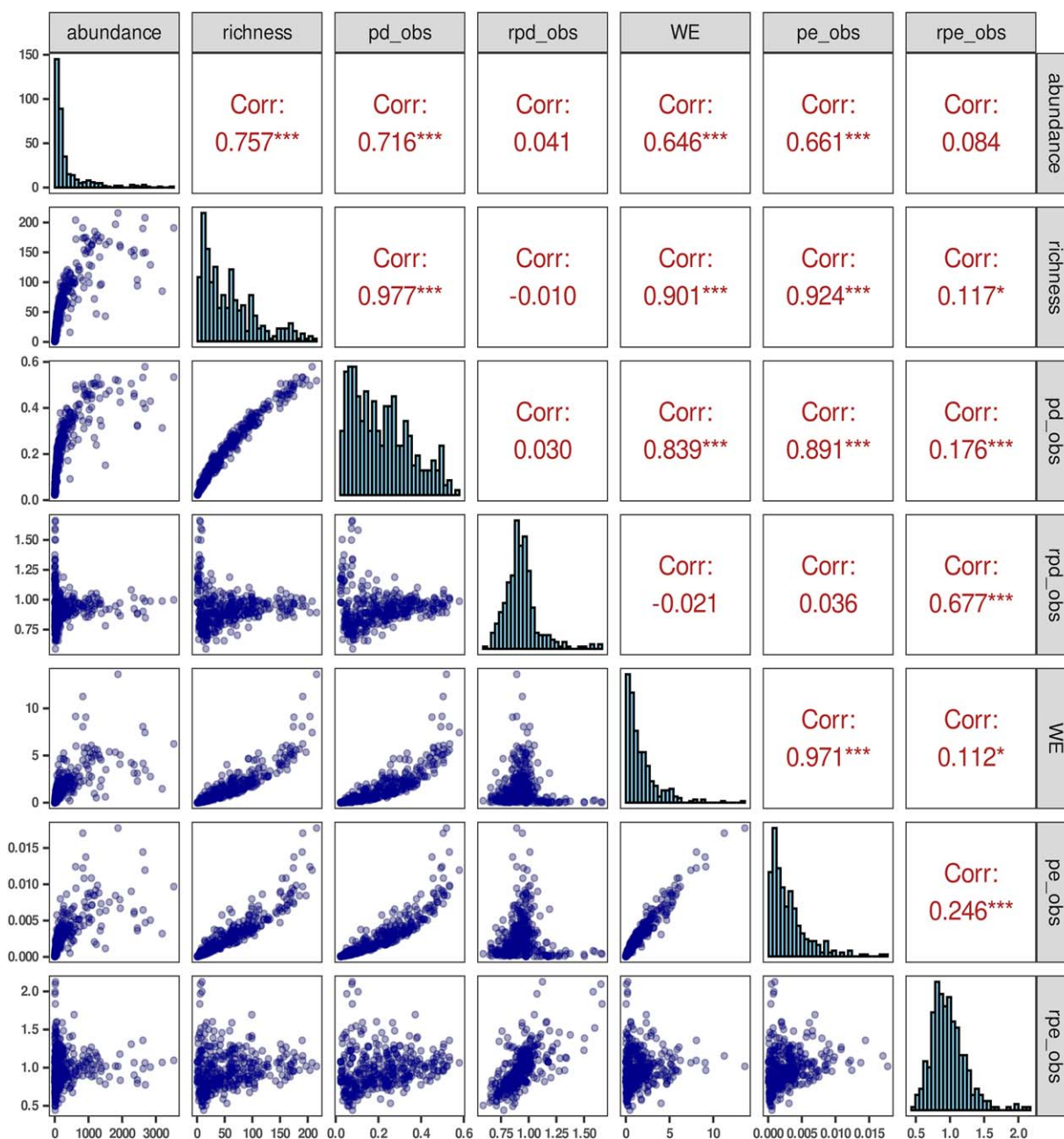


FIG. S3. Pairwise correlation matrix displaying associations between biodiversity indices, including abundance, richness, observed phylogenetic diversity (pd_obs), relative phylogenetic diversity (rpd_obs), weighted endemism (WE), phylogenetic endemism (pe_obs), and relative phylogenetic endemism (rpe_obs) across the Northern Andean páramo. Each scatter plot shows the relationship between pairs of indices, with histograms along the diagonal representing the distribution of each variable. Pearson correlation coefficients are provided in red text above each scatter plot, with significance levels indicated by asterisks (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

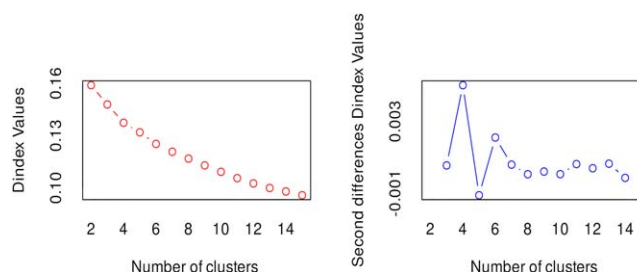


FIG. S4. Diagnostic plots generated from the NbClust analysis using the D index. The left panel shows D index values across a range of cluster numbers, where the optimal number of clusters (k) is indicated by a knee in the curve. The right panel shows the second differences of the D index, with peaks corresponding to these inflection points. Both panels indicate that $k = 4$ is an appropriate selection for the optimal number of clusters.

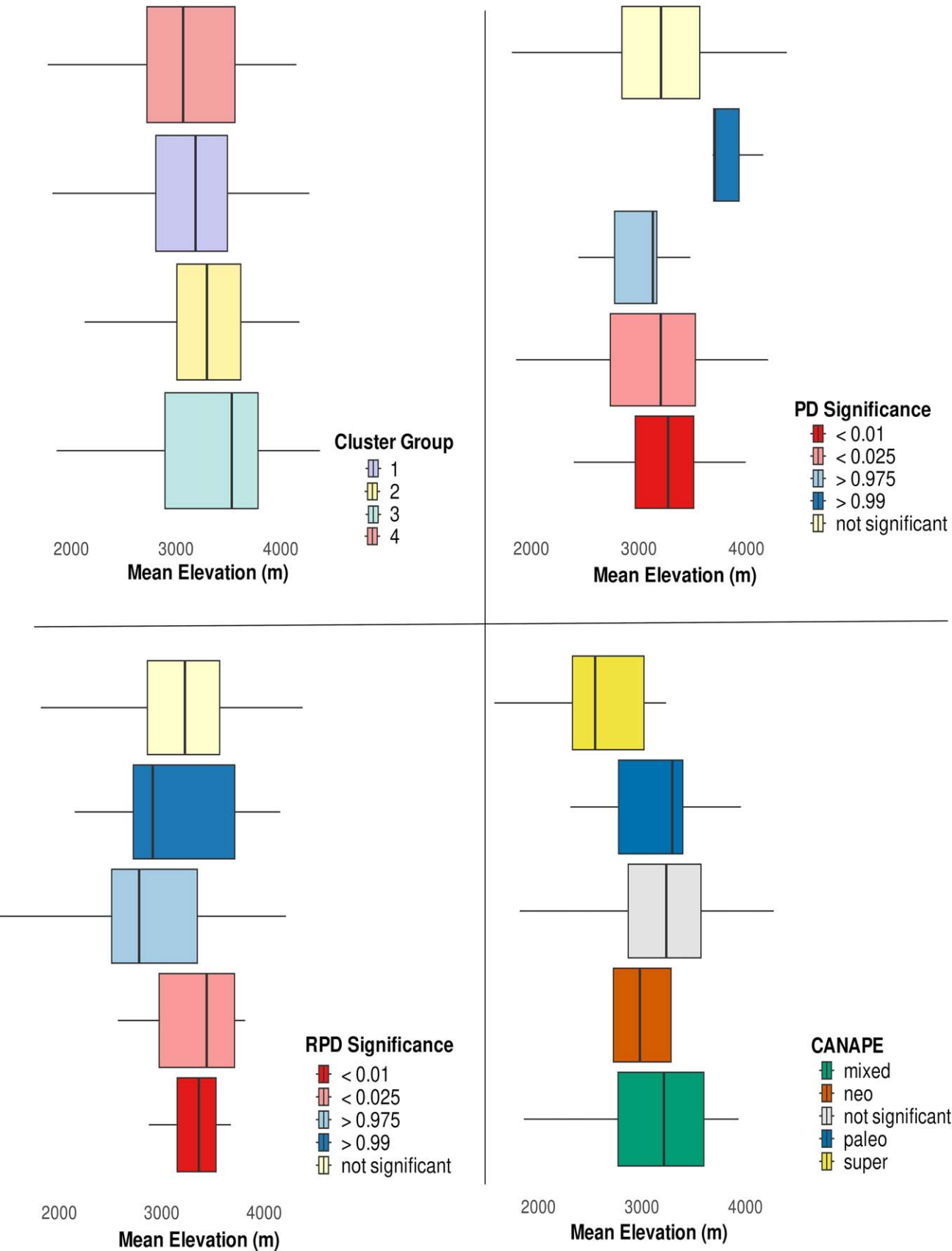


FIG. S5. Box plots illustrating the distribution of mean elevation across different significance levels and cluster groups for various diversity and phylogenetic metrics in páramo regions. A. Cluster groups showing the distribution of mean elevation for each cluster. B. PD significance by elevation. C. RPD significance by elevation. D. CANAPE categories and mean elevation. In the box-and-whisker plots, box widths define the interquartile range, while the vertical line indicates the mean, and the whiskers indicate the data range within 1.5 times the interquartile range.

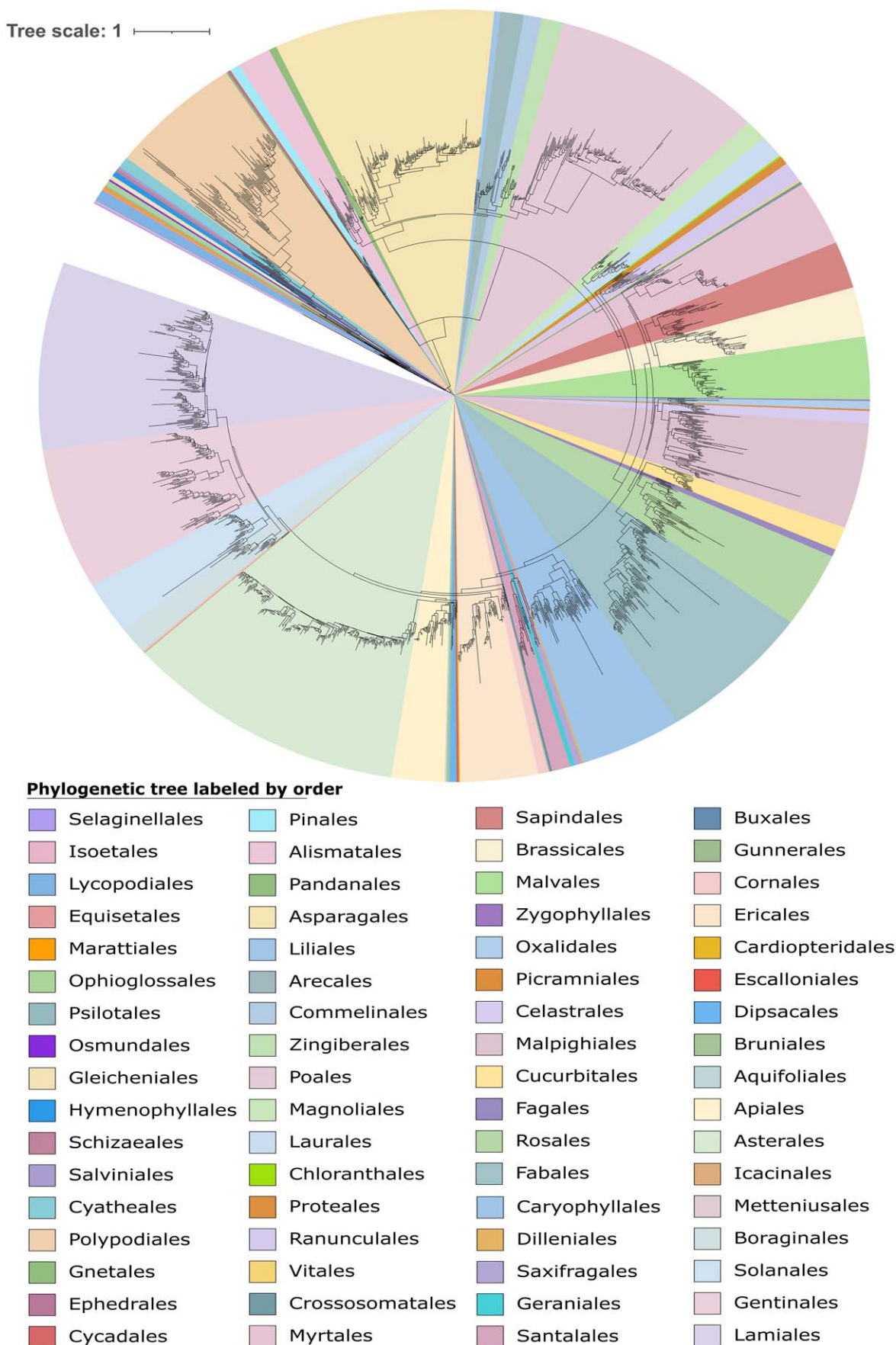


FIG. S6. Phylogeny of páramo vascular plant genera. The 1696 branch tips are coloured to distinguish different orders. Length of the scale bar indicates 1 substitution per site.

TABLE S1. Statistics describing the de novo assemblies of chloroplast genomes from the herbarium specimens. C = complete; S = scaffold; F = failed; Y = yes; N = no; Seq. = sequencing; Herb. = herbarium.

Lab sample identifier	Taxonomic identifier	Source herb. code	Collection number	Seq. yield (Gbp)	Assembly quality	Total assembly length (bp)	No. of contigs	Assembled contigs' seq. depth (X)	Markers annotated?				Sample included in final analysis?
									<i>rbcl</i>	<i>matK</i>	<i>ndhF</i>	<i>trnT</i>	
s1_A	<i>Asplundianthus pseudostuebelii</i>	CUVC	KH01	1.957	C	151,580	1	1345	Y	Y	Y	Y	Y
s2_B	<i>Alloispermum caracasanaum</i>	CUVC	KH03	0.321	F	0	0	0	N	N	N	N	N
s3_C	<i>Agrostopoa wallisii</i>	COL	Barclay 7079	3.735	S	263	1	11	N	N	N	N	N
s4_D	<i>Aristeguietia uribei</i>	COL	Huertás 6778	2.262	S	9581	26	762	Y	N	Y	o	N
s5_E	<i>Aulonemia sp.</i>	COL	Betancur 15470	1.510	C	138,371	1	72	Y	Y	Y	Y	Y
s6_F	<i>Badilloa sp.</i>	COL	Alovar 426	3.823	C	151,761	1	439	Y	Y	Y	Y	Y
s7_G	<i>Josanthus chimborazensis</i>	COL	de Benavides 6941	1.128	S	15,888	35	2808	N	N	Y	N	Y
s8_H	<i>Lourteigia sp.</i>	COL	Fernandez-Alonso 4812	2.331	C	150,913	1	1510	Y	Y	Y	Y	Y
s9_I	<i>Cronquistianthus trianae</i>	COL	de Benavides 9873	3.874	S	44,228	64	113	Y	Y	Y	N	N
s10_J	<i>Selloa ligulata</i>	COL	Cuatrecasas 27882	0.929	S	555	2	226	N	N	N	N	N
s11_K	<i>Ardisia sp.</i>	COL	Betancur 21149	8.272	S	157,758	1	1790	Y	Y	Y	Y	Y
s13_A2	<i>Cybianthus sp.</i>	COL	Betancur 21573	3.435	C	155,454	1	372	Y	Y	Y	Y	Y
s14_B2	<i>Dalea coerulea</i>	CUVC	KH09	3.442	S	4511	13	0	N	N	N	N	N
s15_C2	<i>Daphnopsis sp.</i>	COL	Betancur 23745	2.377	C	173,705	1	276	Y	Y	Y	Y	Y
s16_D2	<i>Dendrophorbium munchiquense</i>	CUVC	KH10	0.953	S	284	1	13595	N	N	Y	N	N
s17_E2	<i>Dicliptera haughtii</i>	COL	Wood 5240	4.205	C	150,619	1	1193	Y	Y	Y	Y	Y
s18_F2	<i>Ditassa caucana</i>	CUVC	KH11	2.484	C	162,866	1	183	Y	Y	Y	Y	Y
s19_G2	<i>Ecrenocarpus viridis</i>	CUVC	KH12	2.637	S	151,840	4	16682	Y	Y	Y	Y	Y
s20_H2	<i>Elymus cordilleranus</i>	CUVC	KH13	14.888	S	114,377	11	1	Y	Y	Y	Y	Y
s21_I2	<i>Epilobium denticulatum</i>	COL	Hernandez 188	2.379	S	136,583	28	328	Y	Y	Y	Y	Y
s22_J2	<i>Galinthe bogotensis</i>	COL	Wesenbeck 119	8.317	S	125,018	11	15	Y	Y	Y	Y	Y
s23_K2	<i>Gaylussacia buxifolia</i>	CUVC	KH14	3.421	S	9359	23	4	N	N	N	N	N
s24_L2	<i>Holodiscus sp.</i>	COL	Betancur 22410	5.262	C	157,433	1	789	Y	Y	Y	Y	Y
s25_M2	<i>Lamourouxia virgata</i>	CUVC	KH15	2.311	S	176,164	6	139	Y	Y	Y	Y	Y
s26_N2	<i>Lippia sp.</i>	COL	Betancur 22614	1.117	C	152,042	1	338	Y	Y	Y	Y	Y
s27_O2	<i>Melica scabra</i>	COL	Glady's Cardenas sn	1.152	F	0	0	0	N	N	N	N	N
s28_P2	<i>Sabazia trianae</i>	COL	Hernandez-Schmidt 879	1.905	C	151,846	1	205	Y	Y	Y	Y	Y
s29_Q2	<i>Xyris subulata</i>	COL	Betancur 23473	2.864	S	148,854	2	988	Y	Y	Y	Y	Y
s30_R2	<i>Diogenesia antioquiensis</i>	COL	Pedraza 2136	6.651	S	133,074	11	123	Y	Y	Y	Y	Y
s31_S2	<i>Evolvulus sp.</i>	COL	Bernal 3200	2.114	S	122,703	13	824	Y	Y	Y	Y	Y
s33_U2	<i>Lepidaploa karstenii</i>	COL	Funk 12005	2.205	C	152,706	1	486	Y	Y	Y	Y	Y
s34_V2	<i>Manettia sp.</i>	COL	Betancur 13030	2.173	C	152,883	1	604	Y	Y	Y	Y	Y
s35_W2	<i>Mikania sp.</i>	COL	Betancur 22272	1.141	C	152,068	1	68	Y	Y	Y	Y	Y

TABLE S2. List of the 567 genera examined in this study, with their corresponding order and family. The GBIF taxon key represents the unique identifier assigned to each genus in the Global Biodiversity Information Facility (GBIF) database.

Count	GBIF taxon key	Order	Family	Genus
1	2873249	Alismatales	Araceae	<i>Anthurium</i>
2	2867589	Alismatales	Araceae	<i>Lemna</i>
3	2756560	Alismatales	Potamogetonaceae	<i>Potamogeton</i>
4	5329186	Alismatales	Juncaginaceae	<i>Triglochin</i>
5	3034812	Apiales	Apiaceae	<i>Apium</i>
6	3034508	Apiales	Apiaceae	<i>Arracacia</i>
7	7508025	Apiales	Apiaceae	<i>Azorella</i>
8	6026305	Apiales	Apiaceae	<i>Bowlesia</i>
9	5371894	Apiales	Apiaceae	<i>Cyclospermum</i>
10	8063199	Apiales	Apiaceae	<i>Daucus</i>
11	3034387	Apiales	Apiaceae	<i>Eryngium</i>
12	3637791	Apiales	Araliaceae	<i>Hydrocotyle</i>
13	3034179	Apiales	Apiaceae	<i>Lilaeopsis</i>
14	3629001	Apiales	Apiaceae	<i>Myrrhidendron</i>
15	3628777	Apiales	Apiaceae	<i>Neonelsonia</i>
16	6027394	Apiales	Apiaceae	<i>Niphogeton</i>
17	3036597	Apiales	Araliaceae	<i>Oreopanax</i>
18	6026743	Apiales	Apiaceae	<i>Ottoa</i>
19	3633853	Apiales	Apiaceae	<i>Perissocoeleum</i>
20	3639035	Apiales	Apiaceae	<i>Sanicula</i>
21	3037700	Apiales	Araliaceae	<i>Schefflera</i>
22	7315729	Aquifoliales	Aquifoliaceae	<i>Ilex</i>
23	2731844	Arecales	Arecaceae	<i>Ceroxylon</i>
24	8236477	Arecales	Arecaceae	<i>Geonoma</i>
25	2850872	Asparagales	Orchidaceae	<i>Aa</i>
26	2839450	Asparagales	Orchidaceae	<i>Acianthera</i>
27	2767031	Asparagales	Asparagaceae	<i>Agave</i>
28	2850864	Asparagales	Orchidaceae	<i>Altensteinia</i>
29	2787512	Asparagales	Orchidaceae	<i>Anathallis</i>
30	10726757	Asparagales	Orchidaceae	<i>Andinia</i>
31	2785169	Asparagales	Orchidaceae	<i>Barbosella</i>
32	2798905	Asparagales	Orchidaceae	<i>Caucaea</i>
33	2790611	Asparagales	Orchidaceae	<i>Coccineorchis</i>
34	2796000	Asparagales	Orchidaceae	<i>Comparettia</i>
35	2789722	Asparagales	Orchidaceae	<i>Cranichis</i>
36	2807675	Asparagales	Orchidaceae	<i>Cyclopogon</i>
37	2832887	Asparagales	Orchidaceae	<i>Cyrtorchilum</i>
38	2832459	Asparagales	Orchidaceae	<i>Draconanthes</i>
39	2767783	Asparagales	Asparagaceae	<i>Echeandia</i>
40	2839230	Asparagales	Orchidaceae	<i>Elleanthus</i>
41	5321019	Asparagales	Orchidaceae	<i>Epidendrum</i>
42	2837096	Asparagales	Orchidaceae	<i>Eurystyles</i>
43	5308859	Asparagales	Orchidaceae	<i>Fernandezia</i>
44	2852506	Asparagales	Orchidaceae	<i>Fronitaria</i>
45	2816918	Asparagales	Orchidaceae	<i>Gomphichis</i>
46	2828306	Asparagales	Orchidaceae	<i>Habenaria</i>
47	2813135	Asparagales	Orchidaceae	<i>Kreodanthus</i>
48	2851992	Asparagales	Orchidaceae	<i>Lepanthes</i>
49	5296843	Asparagales	Iridaceae	<i>Libertia</i>
50	5322840	Asparagales	Orchidaceae	<i>Liparis</i>
51	2808262	Asparagales	Orchidaceae	<i>Malaxis</i>
52	2848856	Asparagales	Orchidaceae	<i>Masdevallia</i>
53	5318055	Asparagales	Orchidaceae	<i>Maxillaria</i>
54	2801835	Asparagales	Orchidaceae	<i>Myoxanthus</i>
55	7947306	Asparagales	Orchidaceae	<i>Myrosmodus</i>
56	5297527	Asparagales	Iridaceae	<i>Olsynium</i>
57	5826372	Asparagales	Orchidaceae	<i>Oncidium</i>
58	2746374	Asparagales	Iridaceae	<i>Orthrosanthus</i>
59	2853981	Asparagales	Amaryllidaceae	<i>Phaedranassa</i>
60	2790726	Asparagales	Orchidaceae	<i>Platystele</i>
61	2787699	Asparagales	Orchidaceae	<i>Pleurothallis</i>
62	2801947	Asparagales	Orchidaceae	<i>Pleurothallopsis</i>
63	2800729	Asparagales	Orchidaceae	<i>Ponthieva</i>
64	2836214	Asparagales	Orchidaceae	<i>Porroglossum</i>
65	5312296	Asparagales	Orchidaceae	<i>Prescottia</i>
66	2790831	Asparagales	Orchidaceae	<i>Pseudocentrum</i>
67	2850741	Asparagales	Orchidaceae	<i>Pterichis</i>
68	2817249	Asparagales	Orchidaceae	<i>Restrepia</i>
69	2849231	Asparagales	Orchidaceae	<i>Scaphosepalum</i>

(Continued)

TABLE S2. (CONTINUED).

Count	GBIF taxon key	Order	Family	Genus
70	2847071	Asparagales	Orchidaceae	<i>Scaphyglottis</i>
71	2816852	Asparagales	Orchidaceae	<i>Sertifera</i>
72	5296234	Asparagales	Iridaceae	<i>Sisyrinchium</i>
73	2787619	Asparagales	Orchidaceae	<i>Specklinia</i>
74	5309193	Asparagales	Orchidaceae	<i>Stelis</i>
75	2859401	Asparagales	Amaryllidaceae	<i>Stenomesson</i>
76	2785234	Asparagales	Orchidaceae	<i>Stenorhynchos</i>
77	2790864	Asparagales	Orchidaceae	<i>Telipogon</i>
78	2791585	Asparagales	Orchidaceae	<i>Trichosalpinx</i>
79	7994043	Asterales	Asteraceae	<i>Achyrocline</i>
80	7764312	Asterales	Asteraceae	<i>Acmella</i>
81	3092067	Asterales	Asteraceae	<i>Aequatorium</i>
82	3099461	Asterales	Asteraceae	<i>Aetheolaena</i>
83	5400249	Asterales	Asteraceae	<i>Ageratina</i>
84	5401760	Asterales	Asteraceae	<i>Ageratum</i>
85	3101273	Asterales	Asteraceae	<i>Alloispermum</i>
86	8424273	Asterales	Asteraceae	<i>Ambrosia</i>
87	3122091	Asterales	Asteraceae	<i>Aphanactis</i>
88	3140635	Asterales	Asteraceae	<i>Asplundianthus</i>
89	8134665	Asterales	Asteraceae	<i>Baccharis</i>
90	3110836	Asterales	Asteraceae	<i>Badilloa</i>
91	7414794	Asterales	Asteraceae	<i>Barnadesia</i>
92	3093425	Asterales	Asteraceae	<i>Belloa</i>
93	7338058	Asterales	Asteraceae	<i>Bidens</i>
94	3089479	Asterales	Asteraceae	<i>Blakiella</i>
95	3144906	Asterales	Asteraceae	<i>Cacosmia</i>
96	3147557	Asterales	Asteraceae	<i>Calea</i>
97	3090235	Asterales	Asteraceae	<i>Chaptalia</i>
98	5405833	Asterales	Asteraceae	<i>Chevreulia</i>
99	4240099	Asterales	Asteraceae	<i>Chionolaena</i>
100	3087713	Asterales	Asteraceae	<i>Chromolaena</i>
101	3105140	Asterales	Asteraceae	<i>Chrysactinium</i>
102	3109505	Asterales	Asteraceae	<i>Chusqueira</i>
103	3104817	Asterales	Asteraceae	<i>Clibadium</i>
104	5397989	Asterales	Asteraceae	<i>Conyza</i>
105	3133912	Asterales	Asteraceae	<i>Coreopsis</i>
106	3112068	Asterales	Asteraceae	<i>Cotula</i>
107	3101576	Asterales	Asteraceae	<i>Critoniopsis</i>
108	3103569	Asterales	Asteraceae	<i>Cronquistianthus</i>
109	3144481	Asterales	Asteraceae	<i>Cuatrecasasiella</i>
110	3108405	Asterales	Asteraceae	<i>Culcitium</i>
111	3098412	Asterales	Asteraceae	<i>Dasyphyllum</i>
112	3109932	Asterales	Asteraceae	<i>Dendrophorbium</i>
113	3090548	Asterales	Asteraceae	<i>Diplostegium</i>
114	3151233	Asterales	Asteraceae	<i>Dorobaea</i>
115	5400009	Asterales	Asteraceae	<i>Erato</i>
116	8199967	Asterales	Asteraceae	<i>Erechtites</i>
117	3144024	Asterales	Asteraceae	<i>Ericentrodea</i>
118	3146961	Asterales	Asteraceae	<i>Erigeron</i>
119	3105074	Asterales	Asteraceae	<i>Espeletia</i>
120	3142374	Asterales	Asteraceae	<i>Eupatorium</i>
121	3130722	Asterales	Asteraceae	<i>Fleischmannia</i>
122	3141097	Asterales	Asteraceae	<i>Floscaldasia</i>
123	8298495	Asterales	Asteraceae	<i>Galinsoga</i>
124	3103209	Asterales	Asteraceae	<i>Gamochaeta</i>
125	5390778	Asterales	Asteraceae	<i>Gnaphalium</i>
126	3138791	Asterales	Asteraceae	<i>Gynoxys</i>
127	5386752	Asterales	Asteraceae	<i>Heliopsis</i>
128	3135221	Asterales	Asteraceae	<i>Hieracium</i>
129	3134192	Asterales	Asteraceae	<i>Hinterhubera</i>
130	3093832	Asterales	Asteraceae	<i>Hypochaeris</i>
131	5386415	Asterales	Asteraceae	<i>Jaegeria</i>
132	3103401	Asterales	Asteraceae	<i>Jalcochila</i>
133	4220446	Asterales	Asteraceae	<i>Joseanthus</i>
134	5402228	Asterales	Asteraceae	<i>Jungia</i>
135	5398060	Asterales	Asteraceae	<i>Laennecia</i>
136	7981606	Asterales	Asteraceae	<i>Laestadia</i>
137	3116921	Asterales	Asteraceae	<i>Lasiocephalus</i>
138	3099893	Asterales	Asteraceae	<i>Lepidaploa</i>
139	3103432	Asterales	Asteraceae	<i>Liabum</i>
140	3122213	Asterales	Asteraceae	<i>Llerasia</i>
141	3087497	Asterales	Asteraceae	<i>Lourteigia</i>

(Continued)

TABLE S2. (CONTINUED).

Count	GBIF taxon key	Order	Family	Genus
142	3092724	Asterales	Asteraceae	<i>Lucilia</i>
143	5697741	Asterales	Asteraceae	<i>Mikania</i>
144	3092559	Asterales	Asteraceae	<i>Monticalia</i>
145	3099401	Asterales	Asteraceae	<i>Munozia</i>
146	3144154	Asterales	Asteraceae	<i>Mutisia</i>
147	3110254	Asterales	Asteraceae	<i>Noticastrum</i>
148	5393253	Asterales	Asteraceae	<i>Oligactis</i>
149	3143176	Asterales	Asteraceae	<i>Onoseris</i>
150	3096721	Asterales	Asteraceae	<i>Oritrophium</i>
151	3112052	Asterales	Asteraceae	<i>Oxylobus</i>
152	3093672	Asterales	Asteraceae	<i>Pappobolus</i>
153	3143084	Asterales	Asteraceae	<i>Paragymoxys</i>
154	3096189	Asterales	Asteraceae	<i>Paranephelus</i>
155	8368631	Asterales	Asteraceae	<i>Pentacalia</i>
156	5391359	Asterales	Asteraceae	<i>Perezia</i>
157	3118688	Asterales	Asteraceae	<i>Philoglossa</i>
158	3088182	Asterales	Asteraceae	<i>Plagiocheilus</i>
159	3103261	Asterales	Asteraceae	<i>Pseudognaphalium</i>
160	3100702	Asterales	Asteraceae	<i>Sabazia</i>
161	3105017	Asterales	Asteraceae	<i>Schistocarpha</i>
162	3101235	Asterales	Asteraceae	<i>Schkuhria</i>
163	3087088	Asterales	Asteraceae	<i>Scrobicaria</i>
164	5385023	Asterales	Asteraceae	<i>Selloa</i>
165	3108482	Asterales	Asteraceae	<i>Senecio</i>
166	3125893	Asterales	Asteraceae	<i>Sigesbeckia</i>
167	3145300	Asterales	Asteraceae	<i>Smallanthus</i>
168	3112131	Asterales	Asteraceae	<i>Soliva</i>
169	3105646	Asterales	Asteraceae	<i>Sonchus</i>
170	3092842	Asterales	Asteraceae	<i>Spilanthes</i>
171	3120449	Asterales	Asteraceae	<i>Steiractinia</i>
172	3125468	Asterales	Asteraceae	<i>Stevia</i>
173	3088468	Asterales	Asteraceae	<i>Tagetes</i>
174	5394163	Asterales	Asteraceae	<i>Taraxacum</i>
175	8346360	Asterales	Asteraceae	<i>Verbesina</i>
176	5939837	Asterales	Asteraceae	<i>Villanova</i>
177	9265090	Boraginales	Cordiaceae	<i>Cordia</i>
178	2925816	Boraginales	Boraginaceae	<i>Cynoglossum</i>
179	7294483	Boraginales	Boraginaceae	<i>Hackelia</i>
180	4065577	Boraginales	Heliotropiaceae	<i>Heliotropium</i>
181	2925752	Boraginales	Boraginaceae	<i>Lappula</i>
182	8099730	Boraginales	Boraginaceae	<i>Lithospermum</i>
183	8207707	Boraginales	Boraginaceae	<i>Moritzia</i>
184	2925668	Boraginales	Boraginaceae	<i>Myosotis</i>
185	4062463	Boraginales	Boraginaceae	<i>Plagiobothrys</i>
186	4064401	Boraginales	Heliotropiaceae	<i>Tournefortia</i>
187	3050478	Brassicales	Brassicaceae	<i>Brayopsis</i>
188	5375388	Brassicales	Brassicaceae	<i>Capsella</i>
189	3045644	Brassicales	Brassicaceae	<i>Cardamine</i>
190	2874482	Brassicales	Caricaceae	<i>Carica</i>
191	3041500	Brassicales	Cleomaceae	<i>Cleome</i>
192	3041482	Brassicales	Brassicaceae	<i>Cremolobus</i>
193	3044104	Brassicales	Brassicaceae	<i>Descurainia</i>
194	3050249	Brassicales	Brassicaceae	<i>Draba</i>
195	5373110	Brassicales	Brassicaceae	<i>Eudema</i>
196	7909561	Brassicales	Brassicaceae	<i>Lepidium</i>
197	3047071	Brassicales	Brassicaceae	<i>Polypsecadium</i>
198	2889934	Brassicales	Tropaeolaceae	<i>Tropaeolum</i>
199	3701750	Bruniales	Columelliaceae	<i>Columellia</i>
200	6388688	Bruniales	Columelliaceae	<i>Desfontainia</i>
201	7323612	Caryophyllales	Amaranthaceae	<i>Alternanthera</i>
202	5384364	Caryophyllales	Amaranthaceae	<i>Amaranthus</i>
203	4160798	Caryophyllales	Basellaceae	<i>Anredera</i>
204	5384489	Caryophyllales	Caryophyllaceae	<i>Arenaria</i>
205	7288905	Caryophyllales	Montiaceae	<i>Calandrinia</i>
206	7741639	Caryophyllales	Caryophyllaceae	<i>Cardionema</i>
207	7267431	Caryophyllales	Caryophyllaceae	<i>Cerastium</i>
208	7327368	Caryophyllales	Nyctaginaceae	<i>Colignonia</i>
209	5588218	Caryophyllales	Caryophyllaceae	<i>Colobanthus</i>
210	5384509	Caryophyllales	Caryophyllaceae	<i>Drymaria</i>
211	3083836	Caryophyllales	Amaranthaceae	<i>Dysphania</i>
212	10679135	Caryophyllales	Amaranthaceae	<i>Iresine</i>
213	9004024	Caryophyllales	Montiaceae	<i>Montia</i>

(Continued)

TABLE S2. (CONTINUED).

Count	GBIF taxon key	Order	Family	Genus
214	4038573	Caryophyllales	Polygonaceae	<i>Muehlenbeckia</i>
215	5528955	Caryophyllales	Caryophyllaceae	<i>Paronychia</i>
216	2889304	Caryophyllales	Polygonaceae	<i>Persicaria</i>
217	4192814	Caryophyllales	Phytolaccaceae	<i>Phytolacca</i>
218	7821030	Caryophyllales	Polygonaceae	<i>Polygonum</i>
219	4035440	Caryophyllales	Polygonaceae	<i>Rumex</i>
220	3085387	Caryophyllales	Caryophyllaceae	<i>Sagina</i>
221	3085897	Caryophyllales	Caryophyllaceae	<i>Silene</i>
222	3085810	Caryophyllales	Caryophyllaceae	<i>Spergula</i>
223	7268258	Caryophyllales	Caryophyllaceae	<i>Stellaria</i>
224	3083518	Caryophyllales	Basellaceae	<i>Ullucus</i>
225	3169095	Celastrales	Celastraceae	<i>Maytenus</i>
226	3086513	Chloranthales	Chloranthaceae	<i>Hedyosmum</i>
227	5303420	Commelinales	Commelinaceae	<i>Callisia</i>
228	7439766	Cornales	Loasaceae	<i>Caiphora</i>
229	3663553	Cornales	Cornaceae	<i>Cornus</i>
230	3645144	Cornales	Hydrangeaceae	<i>Hydrangea</i>
231	7289733	Cornales	Loasaceae	<i>Loasa</i>
232	5646343	Cornales	Loasaceae	<i>Nasa</i>
233	3686247	Crossosomatales	Staphyleaceae	<i>Turpinia</i>
234	7303371	Cucurbitales	Begoniaceae	<i>Begonia</i>
235	3701768	Cucurbitales	Coriariaceae	<i>Coriaria</i>
236	7648864	Cyatheales	Dicksoniaceae	<i>Lophosoria</i>
237	7303679	Cyatheales	Plagiogyriaceae	<i>Plagiogyria</i>
238	3857458	Dipsacales	Viburnaceae	<i>Sambucus</i>
239	4094868	Dipsacales	Caprifoliaceae	<i>Valeriana</i>
240	7272705	Dipsacales	Viburnaceae	<i>Viburnum</i>
241	2653291	Ephedrales	Ephedraceae	<i>Ephedra</i>
242	7287803	Equisetales	Equisetaceae	<i>Equisetum</i>
243	3169458	Ericales	Primulaceae	<i>Ardisia</i>
244	7980642	Ericales	Ericaceae	<i>Cavendishia</i>
245	5682471	Ericales	Ericaceae	<i>Ceratostema</i>
246	3718172	Ericales	Clethraceae	<i>Clethra</i>
247	3722432	Ericales	Primulaceae	<i>Cybianthus</i>
248	4172443	Ericales	Ericaceae	<i>Diogenesia</i>
249	4171314	Ericales	Ericaceae	<i>Disterigma</i>
250	7313906	Ericales	Pentaphylacaceae	<i>Freziera</i>
251	8083690	Ericales	Ericaceae	<i>Gaultheria</i>
252	2882741	Ericales	Ericaceae	<i>Gaylussacia</i>
253	3718860	Ericales	Primulaceae	<i>Geissanthus</i>
254	5528227	Ericales	Theaceae	<i>Gordonia</i>
255	3169308	Ericales	Primulaceae	<i>Lysimachia</i>
256	4170327	Ericales	Ericaceae	<i>Macleania</i>
257	3169412	Ericales	Primulaceae	<i>Myrsine</i>
258	5682809	Ericales	Ericaceae	<i>Orthaea</i>
259	3234143	Ericales	Ericaceae	<i>Plutarchia</i>
260	4173076	Ericales	Ericaceae	<i>Psammisia</i>
261	7328576	Ericales	Ericaceae	<i>Satyria</i>
262	7270807	Ericales	Actinidiaceae	<i>Saurauia</i>
263	4171649	Ericales	Ericaceae	<i>Semiramisia</i>
264	4170953	Ericales	Ericaceae	<i>Sphyrrosporum</i>
265	5541402	Ericales	Styracaceae	<i>Styrax</i>
266	7655338	Ericales	Symplocaceae	<i>Symplocos</i>
267	3189600	Ericales	Pentaphylacaceae	<i>Ternstroemia</i>
268	4168860	Ericales	Ericaceae	<i>Themistoclesia</i>
269	7327778	Ericales	Ericaceae	<i>Thibaudia</i>
270	7971787	Ericales	Ericaceae	<i>Vaccinium</i>
271	2986228	Escalloniales	Escalloniaceae	<i>Escallonia</i>
272	5347113	Fabales	Fabaceae	<i>Astragalus</i>
273	2949607	Fabales	Fabaceae	<i>Chamaecrista</i>
274	2959258	Fabales	Fabaceae	<i>Coursetia</i>
275	5353378	Fabales	Fabaceae	<i>Dalea</i>
276	2967640	Fabales	Fabaceae	<i>Desmodium</i>
277	5356440	Fabales	Fabaceae	<i>Lathyrus</i>
278	2964465	Fabales	Fabaceae	<i>Lupinus</i>
279	2965201	Fabales	Fabaceae	<i>Medicago</i>
280	4924408	Fabales	Fabaceae	<i>Melilotus</i>
281	2969633	Fabales	Fabaceae	<i>Mimosa</i>
282	4051573	Fabales	Polygalaceae	<i>Monnina</i>
283	3191373	Fabales	Polygalaceae	<i>Polygala</i>
284	2957131	Fabales	Fabaceae	<i>Senna</i>
285	8324121	Fabales	Fabaceae	<i>Trifolium</i>

(Continued)

TABLE S2. (CONTINUED).

Count	GBIF taxon key	Order	Family	Genus
286	2975155	Fabales	Fabaceae	<i>Vicia</i>
287	2876131	Fagales	Betulaceae	<i>Alnus</i>
288	3054364	Fagales	Juglandaceae	<i>Juglans</i>
289	3745064	Fagales	Myricaceae	<i>Morella</i>
290	2880642	Fagales	Fagaceae	<i>Quercus</i>
291	2892797	Gentianales	Rubiaceae	<i>Arcytophyllum</i>
292	2901250	Gentianales	Rubiaceae	<i>Cinchona</i>
293	2906930	Gentianales	Rubiaceae	<i>Coccocypselum</i>
294	3170354	Gentianales	Apocynaceae	<i>Cynanchum</i>
295	2908570	Gentianales	Rubiaceae	<i>Diodia</i>
296	3580166	Gentianales	Apocynaceae	<i>Ditassa</i>
297	2898131	Gentianales	Rubiaceae	<i>Faramaea</i>
298	2902194	Gentianales	Rubiaceae	<i>Galianthe</i>
299	2914207	Gentianales	Rubiaceae	<i>Galium</i>
300	7269933	Gentianales	Gentianaceae	<i>Gentiana</i>
301	3169845	Gentianales	Gentianaceae	<i>Gentianella</i>
302	2894616	Gentianales	Rubiaceae	<i>Guettarda</i>
303	5595640	Gentianales	Gentianaceae	<i>Halenia</i>
304	5335451	Gentianales	Rubiaceae	<i>Hoffmannia</i>
305	2908702	Gentianales	Rubiaceae	<i>Ladenbergia</i>
306	3840682	Gentianales	Gentianaceae	<i>Macrocarpaea</i>
307	5338343	Gentianales	Rubiaceae	<i>Manettia</i>
308	2911624	Gentianales	Rubiaceae	<i>Nertera</i>
309	5336799	Gentianales	Rubiaceae	<i>Notopleura</i>
310	2898662	Gentianales	Rubiaceae	<i>Palicourea</i>
311	2919963	Gentianales	Rubiaceae	<i>Psychotria</i>
312	3577476	Gentianales	Apocynaceae	<i>Scyphostelma</i>
313	2918667	Gentianales	Rubiaceae	<i>Spermacoce</i>
314	7269918	Gentianales	Gentianaceae	<i>Symbolanthus</i>
315	9270818	Geraniales	Geraniaceae	<i>Erodium</i>
316	2889949	Geraniales	Geraniaceae	<i>Geranium</i>
317	2651553	Gleicheniales	Gleicheniaceae	<i>Gleichenia</i>
318	3790481	Gleicheniales	Gleicheniaceae	<i>Sticherus</i>
319	3723590	Gunnerales	Gunneraceae	<i>Gunnera</i>
320	7315570	Hymenophyllales	Hymenophyllaceae	<i>Hymenophyllum</i>
321	7423819	Isoetales	Isoetaceae	<i>Isoetes</i>
322	7307804	Lamiales	Lamiaceae	<i>Aegiphila</i>
323	5590767	Lamiales	Gesneriaceae	<i>Alloplectus</i>
324	7869713	Lamiales	Scrophulariaceae	<i>Alonsoa</i>
325	3779399	Lamiales	Acanthaceae	<i>Aphelandra</i>
326	3741986	Lamiales	Plantaginaceae	<i>Aragoa</i>
327	3170806	Lamiales	Orobanchaceae	<i>Bartsia</i>
328	3819520	Lamiales	Gesneriaceae	<i>Besleria</i>
329	4055468	Lamiales	Scrophulariaceae	<i>Buddleja</i>
330	3172125	Lamiales	Calceolariaceae	<i>Calceolaria</i>
331	5372482	Lamiales	Plantaginaceae	<i>Callitriche</i>
332	7332754	Lamiales	Orobanchaceae	<i>Castilleja</i>
333	4085930	Lamiales	Verbenaceae	<i>Citharexylum</i>
334	3886226	Lamiales	Lamiaceae	<i>Clinopodium</i>
335	5590522	Lamiales	Gesneriaceae	<i>Columnnea</i>
336	5575078	Lamiales	Acanthaceae	<i>Dicliptera</i>
337	5414995	Lamiales	Plantaginaceae	<i>Digitalis</i>
338	7297074	Lamiales	Verbenaceae	<i>Duranta</i>
339	4094428	Lamiales	Bignoniaceae	<i>Ecuremocarpos</i>
340	3818976	Lamiales	Gesneriaceae	<i>Glossoloma</i>
341	3822487	Lamiales	Gesneriaceae	<i>Heppiella</i>
342	2926807	Lamiales	Lamiaceae	<i>Hyptis</i>
343	2904597	Lamiales	Acanthaceae	<i>Justicia</i>
344	5592197	Lamiales	Gesneriaceae	<i>Kohleria</i>
345	7829956	Lamiales	Orobanchaceae	<i>Lamoureauxia</i>
346	2925302	Lamiales	Verbenaceae	<i>Lantana</i>
347	2926564	Lamiales	Lamiaceae	<i>Lepechinia</i>
348	3172214	Lamiales	Scrophulariaceae	<i>Limosella</i>
349	7296949	Lamiales	Verbenaceae	<i>Lippia</i>
350	3884495	Lamiales	Lamiaceae	<i>Mimthostachys</i>
351	3898649	Lamiales	Lamiaceae	<i>Obtegomeria</i>
352	3173259	Lamiales	Orobanchaceae	<i>Orobancha</i>
353	3728495	Lamiales	Plantaginaceae	<i>Ourisia</i>
354	3742410	Lamiales	Orobanchaceae	<i>Pedicularis</i>
355	5667828	Lamiales	Lentibulariaceae	<i>Pinguicula</i>
356	8208358	Lamiales	Plantaginaceae	<i>Plantago</i>
357	5341297	Lamiales	Lamiaceae	<i>Prunella</i>

(Continued)

TABLE S2. (CONTINUED).

Count	GBIF taxon key	Order	Family	Genus
358	3905260	Lamiales	Lamiaceae	<i>Salvia</i>
359	2926882	Lamiales	Lamiaceae	<i>Satureja</i>
360	5607886	Lamiales	Lamiaceae	<i>Scutellaria</i>
361	3732571	Lamiales	Plantaginaceae	<i>Sibthorpia</i>
362	2927228	Lamiales	Lamiaceae	<i>Stachys</i>
363	7332183	Lamiales	Plantaginaceae	<i>Stemodia</i>
364	7263313	Lamiales	Acanthaceae	<i>Stenostephanus</i>
365	3172398	Lamiales	Lentibulariaceae	<i>Utricularia</i>
366	7297056	Lamiales	Verbenaceae	<i>Verbena</i>
367	3172077	Lamiales	Plantaginaceae	<i>Veronica</i>
368	3034072	Laurales	Lauraceae	<i>Beilschmiedia</i>
369	5685105	Laurales	Lauraceae	<i>Nectandra</i>
370	7966216	Laurales	Lauraceae	<i>Ocotea</i>
371	3034041	Laurales	Lauraceae	<i>Persea</i>
372	7662198	Laurales	Siparunaceae	<i>Siparuna</i>
373	8271310	Liliales	Alstroemeriaceae	<i>Bomarea</i>
374	2740627	Liliales	Smilacaceae	<i>Smilax</i>
375	2688450	Lycopodiales	Lycopodiaceae	<i>Huperzia</i>
376	2688393	Lycopodiales	Lycopodiaceae	<i>Lycopodium</i>
377	7285964	Magnoliales	Magnoliaceae	<i>Magnolia</i>
378	3878235	Malpighiales	Salicaceae	<i>Abatia</i>
379	5555505	Malpighiales	Clusiaceae	<i>Clusia</i>
380	3059216	Malpighiales	Euphorbiaceae	<i>Croton</i>
381	3073183	Malpighiales	Euphorbiaceae	<i>Dysopsis</i>
382	5646151	Malpighiales	Elatinaceae	<i>Elatine</i>
383	3066321	Malpighiales	Euphorbiaceae	<i>Euphorbia</i>
384	3079083	Malpighiales	Phyllanthaceae	<i>Hieronima</i>
385	7329877	Malpighiales	Hypericaceae	<i>Hypericum</i>
386	7313382	Malpighiales	Passifloraceae	<i>Passiflora</i>
387	5381107	Malpighiales	Phyllanthaceae	<i>Phyllanthus</i>
388	5552863	Malpighiales	Ochnaceae	<i>Sauvagesia</i>
389	7657962	Malpighiales	Violaceae	<i>Viola</i>
390	2874117	Malpighiales	Salicaceae	<i>Xylosma</i>
391	7519667	Malvales	Malvaceae	<i>Acaulimalva</i>
392	7445609	Malvales	Malvaceae	<i>Byttneria</i>
393	3568303	Malvales	Thymelaeaceae	<i>Daphnopsis</i>
394	3935653	Malvales	Malvaceae	<i>Fuertesimalva</i>
395	3934545	Malvales	Malvaceae	<i>Nototriche</i>
396	3152525	Malvales	Malvaceae	<i>Pavonia</i>
397	5617639	Malvales	Malvaceae	<i>Sida</i>
398	5601079	Myrtales	Melastomataceae	<i>Axinaea</i>
399	5600167	Myrtales	Melastomataceae	<i>Blakea</i>
400	3862350	Myrtales	Melastomataceae	<i>Brachyotum</i>
401	3861440	Myrtales	Melastomataceae	<i>Bucquetia</i>
402	3177624	Myrtales	Myrtaceae	<i>Calycolpus</i>
403	3858405	Myrtales	Melastomataceae	<i>Castratella</i>
404	5601663	Myrtales	Melastomataceae	<i>Centronia</i>
405	3869607	Myrtales	Melastomataceae	<i>Chaetolepis</i>
406	3188684	Myrtales	Lythraceae	<i>Cuphea</i>
407	3189067	Myrtales	Onagraceae	<i>Epilobium</i>
408	5418067	Myrtales	Myrtaceae	<i>Eugenia</i>
409	5938223	Myrtales	Onagraceae	<i>Fuchsia</i>
410	5600038	Myrtales	Melastomataceae	<i>Graffenrieda</i>
411	7711707	Myrtales	Melastomataceae	<i>Meriania</i>
412	3867922	Myrtales	Melastomataceae	<i>Miconia</i>
413	3858290	Myrtales	Melastomataceae	<i>Monochaetum</i>
414	3188217	Myrtales	Myrtaceae	<i>Myrcianthes</i>
415	8606669	Myrtales	Myrtaceae	<i>Myrteola</i>
416	3663746	Myrtales	Onagraceae	<i>Oenothera</i>
417	3188517	Myrtales	Melastomataceae	<i>Tibouchina</i>
418	3185061	Myrtales	Myrtaceae	<i>Ugni</i>
419	2650136	Ophioglossales	Ophioglossaceae	<i>Ophioglossum</i>
420	2986074	Oxalidales	Brunelliaceae	<i>Brunellia</i>
421	8427624	Oxalidales	Oxalidaceae	<i>Oxalis</i>
422	4031249	Oxalidales	Elaeocarpaceae	<i>Vallea</i>
423	7333837	Oxalidales	Cunoniaceae	<i>Weinmannia</i>
424	2861235	Pandanales	Cyclocarpaceae	<i>Sphaeradenia</i>
425	5286054	Pinales	Podocarpaceae	<i>Podocarpus</i>
426	2687721	Pinales	Podocarpaceae	<i>Prumnopitys</i>
427	2687731	Pinales	Podocarpaceae	<i>Retrophyllum</i>
428	4140704	Poales	Poaceae	<i>Acicame</i>
429	2706434	Poales	Poaceae	<i>Agrostis</i>

(Continued)

TABLE S2. (CONTINUED).

Count	GBIF taxon key	Order	Family	Genus
430	7747216	Poales	Poaceae	<i>Aira</i>
431	5290083	Poales	Poaceae	<i>Alopecurus</i>
432	4154151	Poales	Poaceae	<i>Andropogon</i>
433	2705975	Poales	Poaceae	<i>Anthoxanthum</i>
434	4106540	Poales	Poaceae	<i>Aulonemia</i>
435	2704103	Poales	Poaceae	<i>Bothriochloa</i>
436	4130399	Poales	Poaceae	<i>Brachypodium</i>
437	2702793	Poales	Poaceae	<i>Briza</i>
438	4120928	Poales	Poaceae	<i>Bromus</i>
439	2719209	Poales	Cyperaceae	<i>Bulbostylis</i>
440	4126206	Poales	Poaceae	<i>Calamagrostis</i>
441	2726565	Poales	Cyperaceae	<i>Carex</i>
442	4152687	Poales	Poaceae	<i>Chusquea</i>
443	4153282	Poales	Poaceae	<i>Cortaderia</i>
444	2730506	Poales	Cyperaceae	<i>Cyperus</i>
445	2705308	Poales	Poaceae	<i>Dactylis</i>
446	2703248	Poales	Poaceae	<i>Danthonia</i>
447	5289416	Poales	Juncaceae	<i>Distichia</i>
448	2716791	Poales	Cyperaceae	<i>Eleocharis</i>
449	5673025	Poales	Poaceae	<i>Elymus</i>
450	4105827	Poales	Poaceae	<i>Eragrostis</i>
451	2690633	Poales	Eriocaulaceae	<i>Eriocaulon</i>
452	2704913	Poales	Poaceae	<i>Festuca</i>
453	2693949	Poales	Bromeliaceae	<i>Greigia</i>
454	2698859	Poales	Bromeliaceae	<i>Guzmania</i>
455	2706164	Poales	Poaceae	<i>Holcus</i>
456	2713131	Poales	Cyperaceae	<i>Isolepis</i>
457	5678417	Poales	Poaceae	<i>Jarava</i>
458	2701617	Poales	Juncaceae	<i>Juncus</i>
459	2704353	Poales	Poaceae	<i>Koeleria</i>
460	4129708	Poales	Poaceae	<i>Lolium</i>
461	2700604	Poales	Juncaceae	<i>Luzula</i>
462	4125611	Poales	Poaceae	<i>Melica</i>
463	4120968	Poales	Poaceae	<i>Muhlenbergia</i>
464	5679387	Poales	Poaceae	<i>Nassella</i>
465	2711900	Poales	Cyperaceae	<i>Oreobolus</i>
466	5287136	Poales	Eriocaulaceae	<i>Paepalanthus</i>
467	2705064	Poales	Poaceae	<i>Panicum</i>
468	4152531	Poales	Poaceae	<i>Paspalum</i>
469	4112352	Poales	Poaceae	<i>Peyritschia</i>
470	5289756	Poales	Poaceae	<i>Phalaris</i>
471	7540661	Poales	Cyperaceae	<i>Phylloscopus</i>
472	6432089	Poales	Poaceae	<i>Piptochaetium</i>
473	5288612	Poales	Bromeliaceae	<i>Pitcairnia</i>
474	4115138	Poales	Poaceae	<i>Poa</i>
475	4122367	Poales	Poaceae	<i>Polypogon</i>
476	2696198	Poales	Bromeliaceae	<i>Puya</i>
477	2694619	Poales	Bromeliaceae	<i>Racinaea</i>
478	2701052	Poales	Cyperaceae	<i>Rhynchospora</i>
479	8303121	Poales	Poaceae	<i>Schizachyrium</i>
480	2711138	Poales	Cyperaceae	<i>Schoenoplectus</i>
481	5289679	Poales	Poaceae	<i>Setaria</i>
482	4149278	Poales	Poaceae	<i>Sporobolus</i>
483	4154714	Poales	Poaceae	<i>Stipa</i>
484	2694745	Poales	Bromeliaceae	<i>Tillandsia</i>
485	4118637	Poales	Poaceae	<i>Triniochloa</i>
486	2704350	Poales	Poaceae	<i>Trisetum</i>
487	5288981	Poales	Bromeliaceae	<i>Vriesea</i>
488	2692845	Poales	Xyridaceae	<i>Xyris</i>
489	7310620	Polypodiales	Aspleniaceae	<i>Asplenium</i>
490	5644076	Polypodiales	Athyriaceae	<i>Athyrium</i>
491	4078397	Polypodiales	Blechnaceae	<i>Blechnum</i>
492	4024925	Polypodiales	Polypodiaceae	<i>Campyloneurum</i>
493	3751948	Polypodiales	Polypodiaceae	<i>Ceradenia</i>
494	8673235	Polypodiales	Pteridaceae	<i>Cheilanthes</i>
495	2650798	Polypodiales	Cystopteridaceae	<i>Cystopteris</i>
496	5275137	Polypodiales	Dryopteridaceae	<i>Dryopteris</i>
497	4011827	Polypodiales	Dryopteridaceae	<i>Elaphoglossum</i>
498	2650378	Polypodiales	Polypodiaceae	<i>Grammitis</i>
499	5532282	Polypodiales	Dennstaedtiaceae	<i>Hypolepis</i>
500	5566260	Polypodiales	Pteridaceae	<i>Jamesonia</i>
501	5566998	Polypodiales	Polypodiaceae	<i>Melpomene</i>

(Continued)

TABLE S2. (CONTINUED).

Count	GBIF taxon key	Order	Family	Genus
502	4030183	Polypodiales	Polypodiaceae	<i>Niphidium</i>
503	2651799	Polypodiales	Pteridaceae	<i>Pityrogramma</i>
504	2650111	Polypodiales	Polypodiaceae	<i>Polypodium</i>
505	7774122	Polypodiales	Dryopteridaceae	<i>Polystichum</i>
506	5275016	Polypodiales	Dennstaedtiaceae	<i>Pteridium</i>
507	3229333	Polypodiales	Polypodiaceae	<i>Terpsichore</i>
508	5937466	Polypodiales	Thelypteridaceae	<i>Thelypteris</i>
509	7287163	Proteales	Proteaceae	<i>Lomatia</i>
510	7287732	Proteales	Proteaceae	<i>Oreocallis</i>
511	5638165	Proteales	Proteaceae	<i>Panopsis</i>
512	7287672	Proteales	Proteaceae	<i>Roupala</i>
513	3033263	Ranunculales	Ranunculaceae	<i>Anemone</i>
514	3982204	Ranunculales	Berberidaceae	<i>Berberis</i>
515	2888374	Ranunculales	Papaveraceae	<i>Bocconia</i>
516	3920944	Ranunculales	Ranunculaceae	<i>Clematis</i>
517	8590528	Ranunculales	Ranunculaceae	<i>Ranunculus</i>
518	3926390	Ranunculales	Ranunculaceae	<i>Thalictrum</i>
519	5370941	Rosales	Rosaceae	<i>Acneta</i>
520	3025344	Rosales	Rosaceae	<i>Alchemilla</i>
521	2984344	Rosales	Urticaceae	<i>Boehmeria</i>
522	5572226	Rosales	Moraceae	<i>Ficus</i>
523	3029817	Rosales	Rosaceae	<i>Fragaria</i>
524	3874545	Rosales	Rhamnaceae	<i>Frangula</i>
525	5369690	Rosales	Rosaceae	<i>Geum</i>
526	2988522	Rosales	Rosaceae	<i>Hesperomeles</i>
527	2987810	Rosales	Rosaceae	<i>Holodiscus</i>
528	3025389	Rosales	Rosaceae	<i>Lachemilla</i>
529	4160094	Rosales	Cannabaceae	<i>Lozanella</i>
530	3030248	Rosales	Rosaceae	<i>Margyricarpus</i>
531	5361795	Rosales	Urticaceae	<i>Parietaria</i>
532	2984417	Rosales	Urticaceae	<i>Pilea</i>
533	5370631	Rosales	Rosaceae	<i>Polylepis</i>
534	8079058	Rosales	Rosaceae	<i>Potentilla</i>
535	3020559	Rosales	Rosaceae	<i>Prunus</i>
536	3039471	Rosales	Rhamnaceae	<i>Rhamnus</i>
537	7577984	Rosales	Rosaceae	<i>Rubus</i>
538	5670265	Rosales	Urticaceae	<i>Urtica</i>
539	4003577	Santalales	Loranthaceae	<i>Aetanthus</i>
540	3990596	Santalales	Santalaceae	<i>Antidaphne</i>
541	7526272	Santalales	Viscaceae	<i>Dendrophthora</i>
542	7288046	Santalales	Loranthaceae	<i>Gaiadendron</i>
543	2889866	Santalales	Viscaceae	<i>Phoradendron</i>
544	4003911	Santalales	Loranthaceae	<i>Tristerix</i>
545	5544309	Sapindales	Anacardiaceae	<i>Mauria</i>
546	3851248	Sapindales	Meliaceae	<i>Ruarea</i>
547	3834860	Sapindales	Rutaceae	<i>Zanthoxylum</i>
548	5689545	Saxifragales	Crassulaceae	<i>Crassula</i>
549	7334214	Saxifragales	Crassulaceae	<i>Echeveria</i>
550	5341760	Solanales	Solanaceae	<i>Brugmansia</i>
551	3801719	Solanales	Solanaceae	<i>Cestrum</i>
552	8285329	Solanales	Convolvulaceae	<i>Cuscuta</i>
553	3800878	Solanales	Solanaceae	<i>Deprea</i>
554	2928672	Solanales	Convolvulaceae	<i>Evolvulus</i>
555	6402978	Solanales	Solanaceae	<i>Ichroma</i>
556	2928509	Solanales	Convolvulaceae	<i>Ipomoea</i>
557	3801410	Solanales	Solanaceae	<i>Jaltomata</i>
558	2928971	Solanales	Solanaceae	<i>Lycianthes</i>
559	2928774	Solanales	Solanaceae	<i>Nicotiana</i>
560	2928990	Solanales	Solanaceae	<i>Nierembergia</i>
561	5341784	Solanales	Solanaceae	<i>Physalis</i>
562	7498435	Solanales	Solanaceae	<i>Salpichroa</i>
563	3798410	Solanales	Solanaceae	<i>Saracha</i>
564	7299177	Solanales	Solanaceae	<i>Sesaea</i>
565	2932050	Solanales	Solanaceae	<i>Solanum</i>
566	8625966	Solanales	Solanaceae	<i>Witheringia</i>
567	3039162	Vitales	Vitaceae	<i>Cissus</i>

TABLE S3. Indicator genera identified in phylogenetic turnover Cluster 1.

Count	Genus	stat	p-value
1	<i>Espeletia</i>	0.55884747	0.001
2	<i>Brachyotum</i>	0.48082072	0.016
3	<i>Escallonia</i>	0.44672895	0.022
4	<i>Vallea</i>	0.41887308	0.008
5	<i>Oreobolus</i>	0.36095708	0.049
6	<i>Castratella</i>	0.34581152	0.032
7	<i>Moritzia</i>	0.3295219	0.005
8	<i>Lourteigia</i>	0.27320492	0.047
9	<i>Oreocallis</i>	0.26302087	0.037
10	<i>Digitalis</i>	0.2521911	0.044

TABLE S4. Indicator genera identified in phylogenetic turnover Cluster 2.

Count	Genus	stat	p-value
1	<i>Carex</i>	0.53467679	0.005
2	<i>Agrostis</i>	0.52824521	0.004
3	<i>Cerastium</i>	0.51335287	0.001
4	<i>Gentianella</i>	0.5090041	0.017
5	<i>Lachemilla</i>	0.50164689	0.01
6	<i>Elaphoglossum</i>	0.49528127	0.037
7	<i>Lupinus</i>	0.49435787	0.039
8	<i>Luzula</i>	0.4930678	0.003
9	<i>Poa</i>	0.49060944	0.001
10	<i>Senecio</i>	0.48925074	0.018
11	<i>Oritrophium</i>	0.48812386	0.007
12	<i>Azorella</i>	0.48023139	0.002
13	<i>Ranunculus</i>	0.46863912	0.002
14	<i>Gentiana</i>	0.46645989	0.018
15	<i>Jamesonia</i>	0.4629655	0.006
16	<i>Polystichum</i>	0.45206385	0.003
17	<i>Festuca</i>	0.4395108	0.042
18	<i>Draba</i>	0.43645651	0.005
19	<i>Melpomene</i>	0.43470545	0.01
20	<i>Arenaria</i>	0.42211227	0.001
21	<i>Erigeron</i>	0.41332599	0.022
22	<i>Isoetes</i>	0.40468169	0.022
23	<i>Aphanactis</i>	0.39916993	0.001
24	<i>Ourisia</i>	0.39528267	0.002
25	<i>Ophioglossum</i>	0.38454205	0.001
26	<i>Lasiocephalus</i>	0.37933718	0.03
27	<i>Aetheolaena</i>	0.37160524	0.027
28	<i>Pseudognaphalium</i>	0.36904624	0.024
29	<i>Koeleria</i>	0.3677306	0.004
30	<i>Aciachne</i>	0.35034715	0.007
31	<i>Distichia</i>	0.34271551	0.019
32	<i>Alopecurus</i>	0.32481315	0.002
33	<i>Potentilla</i>	0.31004592	0.01
34	<i>Montia</i>	0.30240527	0.036
35	<i>Callitriche</i>	0.28290414	0.01
36	<i>Floscaldasia</i>	0.23584127	0.025
37	<i>Lemna</i>	0.23437679	0.008

TABLE S5. Indicator genera identified in phylogenetic turnover Cluster 3.

Count	Genus	Statistic	p-value
1	<i>Solanum</i>	0.52111202	0.013
2	<i>Calceolaria</i>	0.49051045	0.009
3	<i>Bidens</i>	0.4362179	0.045
4	<i>Muehlenbeckia</i>	0.41563525	0.005
5	<i>Clinopodium</i>	0.40859506	0.014
6	<i>Chuquiraga</i>	0.40842414	0.02
7	<i>Salvia</i>	0.40506428	0.017
8	<i>Perezia</i>	0.40327722	0.001
9	<i>Ephedra</i>	0.39985353	0.003
10	<i>Vicia</i>	0.35978888	0.041
11	<i>Ambrosia</i>	0.34416783	0.001
12	<i>Urtica</i>	0.33654135	0.026
13	<i>Alternanthera</i>	0.3293777	0.012
14	<i>Nasa</i>	0.32100334	0.025
15	<i>Silene</i>	0.31322108	0.006
16	<i>Tagetes</i>	0.31074901	0.002
17	<i>Nassella</i>	0.30854497	0.037
18	<i>Jarava</i>	0.30761843	0.003
19	<i>Dalea</i>	0.30448064	0.017
20	<i>Minthostachys</i>	0.29630517	0.012
21	<i>Cheilanthes</i>	0.2939226	0.001
22	<i>Verbesina</i>	0.28661262	0.05
23	<i>Verbena</i>	0.28465185	0.01
24	<i>Caipophora</i>	0.28303618	0.016
25	<i>Stipa</i>	0.28067861	0.026
26	<i>Lamoureauxia</i>	0.27863689	0.045
27	<i>Dysphania</i>	0.27472113	0.001
28	<i>Descurainia</i>	0.27206909	0.011
29	<i>Paranephelium</i>	0.26820821	0.005
30	<i>Altensteinia</i>	0.26545101	0.025
31	<i>Margyricarpus</i>	0.25708674	0.012
32	<i>Pappobolus</i>	0.25651091	0.01
33	<i>Amaranthus</i>	0.25532121	0.003
34	<i>Senna</i>	0.25085936	0.047
35	<i>Erodium</i>	0.24982999	0.013
36	<i>Colignonia</i>	0.24073465	0.037
37	<i>Aira</i>	0.23214561	0.037
38	<i>Smallanthus</i>	0.22916786	0.048
39	<i>Evolvulus</i>	0.21864611	0.03
40	<i>Agave</i>	0.20462732	0.042
41	<i>Polypsecadium</i>	0.19425717	0.037
42	<i>Cardionema</i>	0.19425717	0.035

TABLE S6. Indicator genera identified in phylogenetic turnover Cluster 4.

Count	Genus	Statistic	p-value
1	<i>Anthurium</i>	0.6895429	0.001
2	<i>Begonia</i>	0.67835386	0.001
4	<i>Palicourea</i>	0.6396247	0.001
5	<i>Geonoma</i>	0.56395247	0.001
6	<i>Chamaedorea</i>	0.5312386	0.001
7	<i>Clusia</i>	0.5291601	0.001
8	<i>Siparuna</i>	0.5218321	0.001
9	<i>Centropogon</i>	0.51617436	0.001
10	<i>Saurauia</i>	0.515153831	0.001
11	<i>Psychotria</i>	0.5112211	0.001
12	<i>Glossoloma</i>	0.49596823	0.001
13	<i>Besleria</i>	0.4811281	0.001
14	<i>Hedyosmum</i>	0.47962678	0.002
15	<i>Piper</i>	0.47141278	0.001
16	<i>Cybianthus</i>	0.45064277	0.001
17	<i>Blakea</i>	0.44111241	0.001
18	<i>Mikania</i>	0.44097049	0.001
19	<i>Pammisia</i>	0.43707024	0.001
20	<i>Faramea</i>	0.43540544	0.002
21	<i>Miconia</i>	0.4354019	0.001
22	<i>Guzmania</i>	0.42972265	0.001
23	<i>Nectandra</i>	0.42912125	0.001
24	<i>Pleurothallis</i>	0.42482432	0.009
25	<i>Peperomia</i>	0.41812831	0.001
26	<i>Stelis</i>	0.40702288	0.025
27	<i>Columnea</i>	0.40317465	0.001
28	<i>Boehmeria</i>	0.40317465	0.001
29	<i>Hoffmannia</i>	0.4	0.001
30	<i>Aphelandra</i>	0.3897574	0.001
31	<i>Hieronyma</i>	0.38971768	0.001
32	<i>Drymonia</i>	0.38814111	0.001
33	<i>Koeleria</i>	0.38725336	0.001
34	<i>Rohdea</i>	0.38521286	0.001
35	<i>Cyrtopodium</i>	0.3849301	0.001
36	<i>Tillandsia</i>	0.38111116	0.041
37	<i>Elleanthus</i>	0.38001316	0.008
38	<i>Cyrtilla</i>	0.37122111	0.001
39	<i>Schefflera</i>	0.36346223	0.001
40	<i>Ficus</i>	0.35411504	0.002
41	<i>Cnemidaria</i>	0.35208445	0.001
42	<i>Aciachne</i>	0.34545916	0.032
43	<i>Elaphoglossum</i>	0.3416206	0.002
44	<i>Cleistanthus</i>	0.34154955	0.005
45	<i>Racinaea</i>	0.3412306	0.001
46	<i>Graffenrieda</i>	0.33968311	0.001
47	<i>Epidendrum</i>	0.31818115	0.001
48	<i>Stenostephanos</i>	0.31544015	0.001
49	<i>Ocotea</i>	0.30611111	0.002
50	<i>Tropaeolum</i>	0.30439844	0.033
51	<i>Macrocarpaea</i>	0.29929152	0.013
52	<i>Ternstroemia</i>	0.293554	0.006
53	<i>Eugenia</i>	0.29199842	0.024
54	<i>Notopleura</i>	0.29111112	0.008
55	<i>Alchemilla</i>	0.29111125	0.008
56	<i>Sphryospermum</i>	0.28006112	0.008
57	<i>Styrax</i>	0.28003618	0.008
58	<i>Andropogon</i>	0.27360417	0.002
59	<i>Jaegeria</i>	0.2735506	0.008
60	<i>Hydrangea</i>	0.2673055	0.016
61	<i>Critoniopsis</i>	0.2655325	0.01
62	<i>Croton</i>	0.2634842	0.01
63	<i>Utricularia</i>	0.26335211	0.007
64	<i>Diogenesia</i>	0.26217009	0.007
65	<i>Parietaria</i>	0.2597184	0.016
66	<i>Alloplectus</i>	0.2581989	0.017
67	<i>Nephrolepis</i>	0.2561226	0.029
68	<i>Pitcairnia</i>	0.25544888	0.017
69	<i>Renalmia</i>	0.25511111	0.016
70	<i>Symbolanthus</i>	0.25264558	0.016
71	<i>Myrsine</i>	0.24912222	0.019
72	<i>Galinsoga</i>	0.24839414	0.02
73	<i>Spermacoce</i>	0.24797805	0.02
74	<i>Rynchospora</i>	0.24797802	0.05

(Continued)

TABLE S6. (CONTINUED).

Count	Genus	Statistic	p-value
75	<i>Acalypha</i>	0.24680245	0.019
76	<i>Nephrolepis</i>	0.24151111	0.028
77	<i>Frangula</i>	0.23570226	0.006
78	<i>Parietaria</i>	0.2351226	0.011
79	<i>Zanthoxylum</i>	0.2331262	0.039
80	<i>Attalea</i>	0.22595126	0.039
81	<i>Desmodium</i>	0.22188652	0.039
82	<i>Xylosma</i>	0.21186614	0.035
83	<i>Guettarda</i>	0.19611614	0.035
84	<i>Justicia</i>	0.19611614	0.035
85	<i>Cyperus</i>	0.19611614	0.035
86	<i>Terpsichore</i>	0.19518001	0.037
87	<i>Campyloneurum</i>	0.18848414	0.037